Implementation Run Ledger — Quality & Efficiency Backlog

Purpose. Single source of truth for executing the backlog in `docs/quality-and-efficiency-improvement-plan.md` under the user's instruction: *"implement each change, test in the live application with a ten-company sample, review the result, keep if it works, roll back if it doesn't, then move to the next."*

Status: AUTHORIZED & IN EXECUTION — on 2026-10-15 the user granted: (1) spend on paid production batches; (2) production deployment of each change; (3) dev-testing is acceptable **provided the dev code is identical to production**; (4) judging/scoring is OK; (5) permission to **combine changes** where sensible. Execution has begun with the free offline foundations group.

Deployment-target verification (2026-10-15, before any build)

Hard constraint from the user: the code a change is tested on must be the **same as production**. Verified via Railway GraphQL (`/tmp/check_commits.cjs`, project `db04e5b1…`):

Environment	Service	Deployed commit	Branch
production (`c1166ef6…`)	app	`64d0b9d8…`	main
production	worker	`64d0b9d8…`	main
sprint-10-preview (`6987d716…`)	app	`64d0b9d8…`	main
sprint-10-preview	worker	`64d0b9d8…`	main

Local repo `HEAD` is also `64d0b9d8…`. **All four services + local == `64d0b9d`.** Therefore a feature branch cut from `64d0b9d` and deployed to dev yields `dev = prod + (only the change under test)`, satisfying the constraint. Note: the earlier-recorded "prod `c1166ef6` / dev `6987d716`" were **environment IDs**, not commit hashes — corrected here.

Deploy topology (decisive): Railway `deploymentTriggers` confirms **all four services auto-deploy from branch `main`** of `ahow/companyiq-v3`. There is no separate dev branch; pushing `main` redeploys prod **and** dev together. Governance model is therefore **prod-centric keep/rollback**: merge a WP to main → push → both envs auto-deploy the identical commit → run paid

batch → keep (main stays) or `git revert` + push to roll back. `dev == prod` is satisfied by construction.

Measurement foundations — MERGED & DEPLOYED (2026-10-15 09:47 UTC)

Combined WP10 (per-stage profiler), WP13-A (per-call cost attribution keyed by request identity/emit-timestamp, not log order), WP08 (immutable replay/dependency manifest) on `feat/measurement-foundations` (`bad9f23`), merged `--no-ff` into main as **`2b91d86`** and pushed. All four Railway services observed **BUILDING on `2b91d86`** immediately after push. Change is **read-only / additive**: pipeline outputs byte-identical; each feature has a default-on kill-switch (`RUN_MANIFEST_ENABLED`, etc.). Verified before merge: 21/21 new unit tests pass; client build passes; diff vs `64d0b9d` = +1793/−1 across 13 files (3 new DB tables via idempotent `CREATE TABLE IF NOT EXISTS`). New read paths: `paid_call_events`, `stage_profile_events`, `run_manifests`.

NEXT (not yet run — requires live supervision): instrumented **baseline** paid batch on the frozen roster (company_ids 9–18) via `POST /api/admin/resume-analysis`, once both envs show `2b91d86` = SUCCESS. This baseline anchors both Lane A (scores) and Lane B (speed/cost). ~35 min for 10 companies; record batch id + commit `2b91d86` here, then read `stage_profile_events` / `paid_call_events` / `run_manifests` + scores.

Post-deploy verification + topology CORRECTION (2026-10-15 10:29 UTC)

Confirmed the merge landed healthy on both environments:

check	result
prod app /worker deploy	**SUCCESS** on commit `328d025` (merge + this ledger)
dev app /worker deploy	**SUCCESS** on commit `328d025`
prod app `/api/health`	`{status:ok, version:3.0.0}`
dev app `/api/health`	`{status:ok, version:3.0.0}`
dev worker process	alive, redis+pg ready, pools idle

CORRECTION to the "`dev == prod` by construction" claim above. That is true for the **code** (both on `328d025`) but **FALSE for runtime behaviour and state**. Direct inspection of Railway variables + serviceInstances shows:

1. **Shared state.** Prod `DATABASE_URL` → `postgres.railway.internal`, dev → `pgbouncer.railway.internal`, but **same user / db name `companyiq_v3` / same password hash** and **same `REDIS_URL`**. The two environments share **one Postgres database and one Redis**. Queue name is the hardcoded string `"analysis"` in both `queue.ts` and

`worker.ts` with **no environment namespacing** → there is exactly **one** BullMQ queue.

2. **Different pipeline config, both live on that one queue.** Both worker pools are currently up and connected as `bullmq-worker` to the shared queue:

knob	prod worker (3 replicas)	dev worker (8 replicas)
`FRAMEWORK_V2_MEASURES_PER_CALL`	8	**1**
`FRAMEWORK_V2_TWO_STEP`	—	**1**
`FRAMEWORK_V2_CONTEXT_EXPAND`	—	**1**
`FRAMEWORK_V2_EVIDENCE_ABSENT`	—	**1**
scoring rule	`SCORING_CANONICAL_RULE_GOVERNS=1`	`SCORING_BASE_RATE_PRIOR=true`
`RECONCILE_ENABLED`	true	**false**
`WORKER_CONCURRENCY`	4	2
`JOB/PIPELINE_TIMEOUT_MS`	2.4M / 2.1M	**3.9M / 3.6M**

3. **Consequence (blocks a clean baseline).** Because both pools pull from the **same** queue, a batch enqueued now would be processed by a **nondeterministic mix** of the canonical (prod) and experimental (dev) pipeline configurations. That is not a valid, attributable baseline and would waste the authorized spend. A clean run requires exactly **one** worker pool consuming the queue for its duration (scale the other to 0 replicas).

4. **Strategic implication.** The dev environment already has several Lane-A-style quality knobs enabled as env flags (two-step, context-expand, evidence-absent, base-rate prior, per-call=1). Part of the backlog may already be live-as-flags on dev, which must be reconciled with the “implement one WP at a time from a clean baseline” protocol.

PAUSED for a decision (money + live-infra change + contradicting evidence): which configuration is the baseline/target, and which single worker pool should own the batch runs. No spend and no replica change made unilaterally.

Topology decision RESOLVED — dev aligned to prod + dev worker off the queue (2026-10-15 12:11 UTC)

User decision: production is canonical (`companyiq.online` confirmed = prod custom domain). Dev (sprint-10-preview) settings are to match production exactly, and the dev worker pool is authorized to come off the shared `"analysis"` queue so prod is the sole consumer.

Actions taken (Railway GraphQL, reversible — see `dev_env_rollback_2026-10-15T1210.json`):

1. **Dev worker taken off the queue.** Railway rejects `numReplicas:0` (min 1), and `worker-main.ts` calls `startWorker()` unconditionally (no env guard). So the dev worker `startCommand` was overridden to `/usr/bin/tini -g -- sleep infinity` and `numReplicas` dropped `8 → 1`. The BullMQ consumer never registers → the dev pool no longer pulls jobs. Prod worker unchanged (3 replicas, real start command) = sole consumer.
2. **Dev var alignment (dev → prod values).** Every **non-protected** behavioural / scoring / concurrency / timeout variable on the dev app and dev worker set to its production value; prod-only vars added; dev-only experimental flags removed. Verified: **0 non-protected diffs** vs prod on both services. Key changes:
   - App: `FRAMEWORK_V2_MEASURES_PER_CALL 1→8`, `+SCORING_CANONICAL_RULE_GOVERNS=1`, removed `SCORING_BASE_RATE_PRIOR`, `FRAMEWORK_V2_TWO_STEP/CONTEXT_EXPAND/EVIDENCE_ABSENT`.
   - Worker: `MEASURES_PER_CALL 1→8`, `RECONCILE_ENABLED false→true`, `WORKER_CONCURRENCY 2→4`, `JOB/PIPELINE_TIMEOUT_MS 3.9M/3.6M→2.4M/2.1M`, `PG_POOL_MAX 5→20`, added isolation / adaptive-concurrency / discovery / issuer-sweep / firecrawl / llm-usage-logging vars, removed the four experimental flags.
3. **Protected (left env-specific, NOT copied):** `DATABASE_URL`, `REDIS_URL`, `PORT`, `PG_CONNECTION_TIMEOUT_MS`, all `RAILWAY_*`, all `*_API_KEY` / `SESSION_SECRET` / proxy creds.

Post-change verification: dev app redeploy **SUCCESS** + HTTP 200; dev worker redeploy **SUCCESS** (sleep cmd, 1 replica); `companyiq.online` (prod) HTTP 200; prod worker untouched.

Rollback: restore dev worker `startCommand` + `numReplicas=8`, re-add/delete vars per the rollback JSON. No prod changes were made.

Authorization gate (one-time, blocking)

Each live 10-company production run costs ≈ **$4+ model spend alone** (≈$0.405/co ×10, evidence E2) plus search/proxy/compute. Across ~7 behavioural WPs, each with an implement-run and a possible rollback re-run, the full sequence is **~15–20 paid production batches (~$80–150+ model cost)**, and each production deploy is an irreversible change to the live service. The plan document flags production/paid/ scoring/migration steps as needing **separate approval**.

Required before any paid run: explicit go-ahead on (1) the spend, (2) production deployment of each change, (3) dev-first testing being acceptable. Until then this ledger only accumulates **free, reversible** work (code on a feature branch, offline builds, this ledger, the frozen roster).

Frozen 10-company roster (identical inputs for every WP)

Frozen at (UTC): 2026-10-15 (scaffolding) **Source list:** company_lists.id = 7 (“25 companies”, workspace 3) — same provenance as the adjudicated 25-company validation (VALIDATION25_ROSTER.md) and the authorized 10-company acceptance run (batch 1275). **Selection rule:** first 10 members of list 7 → company_ids **9,10,11,12,13,14,15,16,17,18 Roster id CSV:** `9,10,11,12,13,14,15,16,17,18`

company_id	name
9	CSX Corporation
10	MAGNA INTERNATIONAL INC.
11	THE SOUTHERN COMPANY
12	S&P GLOBAL INC.
13	MARTIN MARIETTA MATERIALS, INC.
14	DARDEN RESTAURANTS, INC.
15	BCE INC. *(E3 unresolved-disclosure diagnostic anchor)*
16	STRYKER CORPORATION
17	AMPHENOL CORPORATION
18	AFLAC INCORPORATED

Workspace 3 · framework 18 (36 measures). Roster is frozen BEFORE any enqueue; launches pass explicit companyIds (provenance list_id=7). Every WP is judged on this identical set.

How “works / doesn’t work” is judged (per category)

Category	WPs	Pass criterion
Behavioural, live-10-co	A2, A4, A5, B1, B3, B5, B6	10-co live sample vs WP01 adjudicated baseline; semantic WPs also need an equivalence/held-out check so a change can’t silently alter answers
Gated	B4 (needs A1+A3), B-dur (needs confirmed clean re-run)	Run only after prerequisites pass
Foundational / offline	A0/WP01, A7/WP08, B0/WP10 profiler, B0/WP13-A	“Works” = builds + emits correct artifacts (no live score)
Governance / reporting	A1, A3, A6, A9	Verified by output/diff
Verify-only	A8/WP09	Confirm already-present behaviour; no rebuild

Pass bar for behavioural live runs (reuse of the batch-1275 acceptance bar): 10/10 first-attempt completions, each ≤ 35 min from claim, any retry voids the pass; AND no accuracy/completeness regression vs baseline. Speed/cost wins that regress accuracy/completeness = **rollback** (hard rule).

Dependency-ordered execution sequence

1. Foundations (offline, free): WP01 baseline harness · WP08 replay keys · WP10 profiler · WP13-A attribution
2. A1 (governance) → A2 (behavioural) → A3 (governance)
3. B1 (behavioural)
4. A4 → A5 (behavioural)
5. B3 → B4 (gated on A1+A3)
6. A6 (governance)
7. B5 (behavioural)
8. B6 (behavioural)
9. A9 (reporting)
10. B-dur (gated on a confirmed clean re-run)

Outcome ledger (filled as each WP runs)

WP	Branch/ commit	Deployed to	Batch id	Result vs baseline	Decision	Rollback note
WP01	328d025 (deployed)	prod (all 4 svc)	1284	BASELINE CAPTURED: 10/10 first-attempt, max 26m42s claim→done, sum_score=156/met=56, Yes=56/No=304, cost $3.95, 13.7M tok	✅ reference established	n/a (reference)
WP08	—	—	—	—	pending	—
WP10	—	—	—	—	pending	—
WP13-A	—	—	—	—	pending	—
A1	e6b2e85 + 08b56ee	prod (all 4 svc)	1285	ADDITIVE/NON-REGRESSIVE: 10/10 first-attempt, max 30m46s claim→done (<35m bar), all ms=36, 360 measure_scores (=baseli	✅ keep	revert e6b2e85 +08b56ee if ever needed (additive cols only)

WP	Branch/ commit	Deployed to	Batch id	Result vs baseline	Decision	Rollback note
				ne), sum_score=152/met=55 & Yes=55/No=305 (within run-to-run variance; baseline 156/56, 56/304). **Cell-flip vs baseline 1284: Y→N=11, N→Y=10, Total Flips=21** (net −1 Yes conceals 21 changed cells — see Comparison methodology below). cost $3.94/13.68M tok (≈baseline $3.95/13.7M).		

WP	Branch/ commit	Deployed to	Batch id	Result vs baseline	Decision	Rollback note
				Code diff (328d025→e6b2e85) touches NO scoring/ analyzer file ⇒ additivity by construction. research Status/extraction_state verified live.		

A1 reuse-path completeness fix (08b56ee): verification found `markLinkedDocumentVerified` (cache-reuse promote path) left `extraction_state` NULL on reused docs — 292/338 fetch=ok docs (77%) unlabeled in batch 1285, breaking A1's "every required source has a traceable outcome" acceptance + coverage denominator. Fixed via scalar-subquery derivation (promotion to 'ok' never gated on the join; mirrors recordFetchSuccess). Live-validated by backfilling batch 1285: all 292 → 'complete', now **338/338 fetch=ok docs (100%) labeled**, remaining 42 NULL = exactly the 19 dead + 23 rejected (never extracted ⇒ correctly NULL). In-pipeline behaviour of the promote-path write will be confirmed incidentally in the next WP's batch (warm corpus ⇒ heavy reuse). | A2 | 7702606 | prod (all 4 svc) | 1286 | EVIDENCE-GATE HARDENED, versioned (v2 default; v1 via `EVIDENCE_GATE_POLICY=v1`). Fixes E6 #1/#2/#3: NFKC Unicode-aware normalisation (preserves digits/−/%/decimals/non-Latin; removed ASCII-only strip), removed fuzzy `lcs≥120` escape hatch + honest-ellipsis span validation, explicit source-attribution failure instead of silent pack-wide fallback, anti-echo override for authoritative provenance. **Correctness proven by 17 unit tests (QI01–QI09 + idempotence/non-mutation/fail-open), legacy gate+provenance tests still green under v1.** Live batch 1286: 10/10 first-attempt, 38m24s, all ms=36 (no measure collapse), docs gate 403 accept/29 reject (6.7%), cost $3.9531/13.72M tok (≈baseline — pure string gate, zero LLM overhead). Aggregate 58 yes vs baseline 56/A1 55 (within run-to-run band). **Cell-flip vs baseline 1284: Y→N=9, N→Y=11, Total Flips=20** (net +2 Yes conceals 20 changed cells;

vs A1 1285: Y→N=7, N→Y=10, Total Flips=17 — see Comparison methodology + table below). **No over-rejection**: three 0/0 cos (MartinMarietta/Darden/Amphenol) are baseline-stable (0/0 in 1284 & 1285 too — see L281), NOT caused by A2. | ✅ keep (v2 on) | revert to v1 via env flag (no redeploy) or revert 7702606 | | A3 | 3990bcd | prod (app + worker, both SUCCESS) | 1287 | ELIGIBILITY COMPOSITION GUARDRAIL added as a standalone module (`server/lib/eligibility-decision.ts`) that carries an **immutable, frozen eligibility decision** (entity · issuer/author · source class · as-of/version · scope) through compose → recompute-per-target (I1) → bounded repair (I2); repair can fill missing provenance but **can never override an exclusion**; applies framework parent/subsidiary scope (`entity_only` vs `group_consolidated`); recomputes per target so reused source bytes can't carry stale eligibility. Versioned kill-switches: `ELIGIBILITY_POLICY` (v2 default = compose+enforce logic available / v1 = legacy passthrough) and `ELIGIBILITY_ENFORCE` (unset/off = **shadow**: compose & attach decision, zero verdict mutation / on = downgrade excluded positives to No). **Correctness proven by 20 unit tests (EP01–EP06 + kill-switch/immutability/totality/serialization), 20/20 pass; legacy suites still green (evidence-gate 14/14, evidence-gate-a2 17/17, provenance-gate 15/15, provenance 40/40).** tsc gate: 61 baseline == 61 current, **zero new diagnostics**. Live batch 1287 ran in **shadow mode** (no `ELIGIBILITY_ENFORCE` in prod env ⇒ zero verdict impact by construction): 10/10 first-attempt, all ms=36 (no measure collapse), cost $3.8945/13.52M tok (≈baseline $3.95/13.7M — checks are string/compute-cheap, zero LLM overhead). **Cell-flip vs baseline 1284: Yes 56→64, Net +8, Y→N=10, N→Y=18, Total Flips=28; vs A2 1286: Yes 58→64, Net +6, Y→N=9, N→Y=15, Total Flips=24** (see tables below). These flips are **run-to-run model variance, not A3-caused** — shadow mode mutates no verdict. **No collapse / over-rejection**: the three all-No cos (Martin Marietta/Darden/Amphenol) stay 0/0 where expected; the only loss (Aflac −3 vs baseline) is already present in A2 (6→3 at 1286) and is stable A2→A3 (3→3), i.e. reproducible corpus variance independent of A3. | ✅ keep — now **enforce-ON** via `15bbcc0` (framework-level provenance fix; VERDICT_CHANGE=0, Y→N=0, N→Y=0, zero regressions — see 2026-10-16 section) | kill switches intact: `ELIGIBILITY_ENFORCE=off` (shadow) / `ELIGIBILITY_POLICY=v1` (legacy passthrough); or revert 3990bcd/15bbcc0 | | B1 | 57716af | prod (all 4 svc, auto-deploy from main; app+worker SUCCESS on 57716af; health `{"status":"ok","version":"3.0.0"}`) | 1293 (corpus-replay off 1288; `pipeline_version=57716af` on all 360 measure_scores rows + manifest gitSha=57716af all 10 co ⇒ worker provably on B1 code) | WP10/B1 prep-stage compute cache (exact-equivalence speedup). New module `server/lib/prep-compute-cache.ts` precompiles the peer-scan regex once, precomputes BM25 term maps/doc-lengths under content keys, skips no-op DB writes, bounds caches; wired into `passage-retrieval.ts`, `aboutness.ts`, `storage.ts`. Versioned kill-switch **`PREP_COMPUTE_CACHE_POLICY`** (`on` DEFAULT = B1 active / `off` = exact pre-B1

behaviour). **Equivalence proven by offline byte-identical guard (authoritative)** over full fixture (104 docs, 4601 chunks, **full 2675-peer universe uncut**, 36 queries): scoreAboutness per-doc, bm25Score per query×chunk, and BM25 ranking+tie-breaks all **identical OFF vs ON ⇒ byte-identical LLM input**. Speedup: BM25 scoring p50 264.9→3.8ms (−98.6%), p95 321.3→7.5ms (−97.7%); peer-scan aboutness p95 536.4→451.2ms (−15.9%) (one-time cost shifts into buildBM25Index). **38/38 tests green (prep-compute-cache 19 new + legacy suites); tsc 0 new diagnostics.** Live batch 1293: 10/10 first-attempt, 0 fail, 10m38s/3 replicas, cost $4.1684/14.57M tok (≈baseline 1288 $4.23/14.74M — byte-identical prompts). **Cell-flip vs baseline 1288: Yes 70→67, Net −3, Y→N=11, N→Y=8, Total Flips=19** (within LLM noise floor A1 21/A2 20/A3 28 — run-to-run model variance, NOT B1; B1 produces byte-identical LLM input per offline guard, and `quotes` is LLM-cited not raw pack). **Total cells Y→N=11, N→Y=8.** | ✅ keep (default `on`) | `PREP_COMPUTE_CACHE_POLICY=off` (no redeploy) or revert 57716af (6-file self-contained module+wiring; no schema/env change) | | A4 | (pending push) | pending deploy | n/a (no batch — verdict-invariant) | WP05/A4 extraction/table/OCR/translation fidelity. Framework-level outcome-classifier resolves the small-PDF runtime gap at ROOT CAUSE (pdf-parse xref rejection ≠ empty doc); recovery-boundary guard preserved by non-interference (B1-1..B1-7 green); subprocess timeout now marks result incomplete (partial stdout never promoted to complete); adaptive per-page OCR (gated on text-layer check, all pages, no 8-page cap); table/translation/boilerplate fidelity asserted. **EX01–EX07 all green (10/10)**, regression suites 12/12 (recovery-boundary B1, equivalence B6, stalled-extraction B3/B2, table-horizon). tsc: 0 new diagnostics in changed files. Verdict-invariant: extraction layer returns `.text` byte-for-byte to all 11 callers; ExtractionFidelity is additive metadata only — scoring untouched. | ✅ keep | revert the A4 commit (processor.ts extraction-layer refactor is self-contained; no schema/env change) | | A5 | ca0939c → on: f15f8b6 | prod (all 4 svc, push 2026-10-16 09:16 UTC; GATE5 flip push 2026-10-16 14:56 UTC; health `{"status":"ok","version":"3.0.0"}`) | 1288 (corpus-replay off 1287; snapshot gitSha=f15f8b6 all 10 co ⇒ worker on new code ⇒ policy=on) | WP06/A5 retrieval & evidence-pack recall. New pure DB-free/LLM-free module `server/lib/evidence-recall.ts`: framework-level decisive-passage oracle (measure `title+evidenceKeywords` only), passage-level recall measurement AT THE ACTUAL PROMPT BUDGET, and a BOUNDED append-only expansion planner that COMPOSES with A3 (injected eligibility predicate ⇒ excluded material never reintroduced), never shrinks top-k/pack length, dedups only when every unique decisive fact+source link stays recoverable, deterministic stable passage id+order. Versioned kill-switch **`RECALL_EXPANSION_POLICY`** (`off` DEFAULT = current prod/byte-identical / `shadow` = telemetry only / `on` = apply expansion); verdict-changing retrieval NEVER default, NOT set in prod env. analyzer.ts wiring strictly gated at the pre-scoring evidence seam — when `off` the block is skipped entirely (no extra work, pack byte-identical). **RT01–RT06 + kill-**

switch-default + no-op-append: 8/8 pass. Free/offline GATE2 recall benchmark over the frozen 13-company fixture corpus (real `summarizeDocuments` BM25, no LLM): 377 (company×measure) cells, **micro-recall 74.2%** (3334/4493 decisive passages recalled), macro 74.3%, 1159 misses of which **537 recoverable** by the bounded shadow expansion. tsc: 0 new diagnostics in changed files (62 pre-existing elsewhere). At the shipped-off default: **Total cells changing Y→N = 0, N→Y = 0** (no scoring surface touched). **GATE5 (2026-10-16, user-approved): default flipped `off→on` via code (`f15f8b6`); env still overrides.** Live corpus-replay batch 1288 off 1287's frozen corpus (1284/1285 corpora pruned ⇒ replay off 1287 instead): 10/10, all ms=36, snapshot gitSha=f15f8b6 on all 10 co confirms policy=on, cost $4.23/14.74M tok. **Cell-flip vs 1287 baseline: Yes 64→70, Net +6, Y→N=5, N→Y=11, Total Flips=16** (bounded expansion adds decisive evidence ⇒ more No→Yes than Yes→No; see 2026-10-16 section). | ✅ keep — now default **on** via `f15f8b6` (GATE5 live batch 1288; Y→N=5, N→Y=11, Total Flips=16, Yes 64→70; see 2026-10-16 section) | `RECALL_EXPANSION_POLICY=off` (or `=shadow`) restores prior behaviour with NO redeploy; or revert `f15f8b6` (3-file default+comments+test, no schema change) then ca0939c for the whole feature | | B3 | c25afd4 | prod (all 4 svc, auto-deploy from main; health `{"status":"ok","version":"3.0.0"}`) | n/a (no batch — equivalence-preserving at shipped-off default) | WP11/B3 discovery dedupe + bounded parallelism. New module `server/lib/discovery-parallel.ts` (content-hash candidate dedupe + bounded-concurrency `webSearchBatch`), integrated into `discovery.ts` and wired into the Generic-Artefact + R6a IR-Platform discovery lanes. Versioned kill-switch **`DISCOVERY_PARALLEL_POLICY`** (`serial` DEFAULT = current prod/byte-identical / `shadow` = run parallel path for telemetry, return serial result / `on` = apply parallel+dedupe) plus knob `DISCOVERY_MAX_PARALLEL` (default 4). **Equivalence proven: same candidate SET and same accumulation order as serial** — dedupe only collapses exact content-hash duplicates, so the discovered corpus is identical; parallelism only reduces wall-clock + near-dup web calls. **DP01–DP10 unit suite 15/15 pass.** tsc: 0 new diagnostics in changed files (62 pre-existing elsewhere). At the shipped-off default (`serial`) the parallel path is skipped entirely ⇒ **Total cells changing Y→N = 0, N→Y = 0, Total Flips = 0** (no scoring surface touched; discovery corpus byte-identical). | ✅ keep (default `serial`/off) | `DISCOVERY_PARALLEL_POLICY=serial` (default; no redeploy needed) or revert c25afd4 (3-file self-contained module + integration; no schema/env change) | | B4 | 08212cd (module) + 957c263 (wiring) | prod (all 4 svc, auto-deploy from main; health `{"status":"ok","version":"3.0.0"}`) | n/a (no batch — equivalence-preserving at shipped-off default) | WP12/B4 layered content-hash cache. **Gate satisfied: B4 needs A1 (e6b2e85+08b56ee, batch 1285 ✅) + A3 (3990bcd+15bbcc0, enforce-on ✅).** New module `server/lib/cache-layers.ts` (7-level layered cache design with sha256 content-hash keys reusing corpus-replay/run-manifest primitives; tenant isolation via `workspaceId` in every key; reason-aware negative caching;

A1/A3 composition; FIFO eviction at `CACHE_LAYER_MAX_ENTRIES=2048`) + wiring layer `server/lib/pipeline-cache.ts` (singleton stores + gated wrappers). Versioned kill-switch **`CACHE_LAYER_POLICY`** (`off` DEFAULT = current prod/byte-identical / `shadow` = compute+compare, return fresh / `on` = serve cache). **Scope — honestly bounded: 2 of 7 layers wired** — (1) **bytes layer** (pipeline.ts fetch phase: caches extracted text by `{workspaceId,url}` ⇒ a warm hit skips network I/O *and* PDF extraction/OCR — the dominant 446s→10.8s lever) and (2) **index layer** (analyzer.ts `summarizeDocuments`: caches BM25-selected corpora). **Deferred (not wired): extraction, OCR, eligibility, packs, scores** — each carries an A4 (extraction-fidelity) / A1 (coverage-labelling) invariant-preservation risk or LLM-output non-determinism risk that would need its own equivalence gate; recorded here so the 2/7 scope is explicit, not implied-complete. **cache-layers 15/15 + cache-wiring integration 12/12 pass** (off-bypass, cold==warm under `on`, shadow-returns-fresh, freshness-keyed-by-content-hash, tenant isolation). tsc: 0 new diagnostics in changed files. At the shipped-off default (`off`) both wrapped seams bypass the cache entirely ⇒ **Total cells changing Y→N = 0, N→Y = 0, Total Flips = 0** (byte-identical to prod). | ✅ keep (default `off`); extraction/OCR/eligibility/packs/scores layers DEFERRED to a future WP extension | `CACHE_LAYER_POLICY=off` (default; no redeploy) or revert 957c263 (wiring) then 08212cd (module); no schema/env change | | A6 | ec8aa4e | prod (all 4 svc, auto-deploy from main; push 7134eb5..ec8aa4e; health `{"status":"ok","version":"3.0.0"}`) | n/a (no batch — governance, verified by output/diff; shadow default = verdict-invariant by construction) | WP07/A6 framework-contract + test-drive defect governance. New module `server/lib/framework-contract.ts` (+ test) that (1) assembles a strict per-measure CONTRACT view (construct · decidable definition · qualifying/non-qualifying evidence · source/entity/time scope · applicability · implementation-vs-aspiration · anti-inference · uncertainty) by **reusing existing framework-v2 modules** rather than duplicating them, (2) classifies test-drive defects into 7 real causes (framework-ambiguity / discovery / acquisition-extraction / entity-provenance / retrieval / scoring / technical) so a missed document routes to a RESEARCH fix not a looser rubric, and (3) enforces the Goodhart no-distribution-tuning rule as a HARD RULE (prevalence can never move a definition or score; accepted edits emit a NEW framework version with an impact preview, never a silent rewrite; company-specific fixes rejected unless a universal data correction). Versioned kill-switch **`FRAMEWORK_CONTRACT_POLICY`** (`shadow` DEFAULT = non-mutating passthrough / `enforce` = NOT set in prod), `FRAMEWORK_CONTRACT_POLICY_VERSION=1`. `runContractGovernance(...).publishedScoresMutated` is **always false ⇒ zero published scores changed by construction**. Reused (not duplicated): scoring-contract.ts, structured-guidance.ts (extractGuidanceObject), anti-inference.ts, rules.ts, eligibility-decision.ts, root-cause-diagnostic.ts, builder-improve.ts, emitHardenedFramework. Genuine gaps closed: G1 assembled contract view + 2 invariants (examples

illustrate, never mutate; search hints ≠ eligibility); G2 7-category defect classifier w/ missed-doc→research routing; G3 no-distribution-tuning validator w/ new-version + impact-preview (ImpactPreview exposes cellsYesToNo/cellsNoToYes/cellsChanged for any future enforce run). **Tests 32/32 (FW01–FW07 = 3/5/3/3/3/2/3 + kill-switch-default 3 + non-mutation 3 + serialization/assembly 6).** tsc: 0 new diagnostics in changed files (62 pre-existing baseline unchanged). **At shipped-off default (`shadow`): Total cells Y→N = 0, N→Y = 0, Total Flips = 0** (no scoring surface mutated). | ✅ keep (default `shadow`) | `FRAMEWORK_CONTRACT_POLICY=shadow` is the default (no redeploy); or revert ec8aa4e (2-file self-contained module+test; no schema/env change) | | B5 | 154ff9b → on: 8725521 | prod (all 4 svc, auto-deploy from main; shipped-dormant push SUCCESS on 154ff9b; **GATE5 flip push 2026-10-17 SUCCESS on 8725521 all 4 svc**; health `{"status":"ok","version":"3.0.0"}`) | 1294 (corpus-replay off 1293's frozen corpus; snapshot gitSha=8725521 all 10 co ⇒ worker on new code ⇒ policy=on; canary-first company 9) | WP13-C/D/E/B5 model-cost reductions (semantic — ships DORMANT). New framework-level module `server/lib/model-cost.ts` (DB-free, no company/topic branches) with three levers: **(C, headline) `planContextTrim`** — trims *repeated* context but **provably never *decisive* context (LM04)**: reuses A5's `computeDecisivePassages` oracle (not reimplemented) and mirrors `planBoundedExpansion`'s subsumption rule in the *trim* direction — a pack passage is a trim candidate ONLY if it is non-decisive AND every decisive token + source link it carries is already covered by a kept passage. HARD INVARIANT (`verifyTrimPreservesRecall`): decisive pack passages ⊆ kept set, recall preserved with identical misses ⇒ relevant/decisive text is NEVER dropped; **(D) `stablePrefixOrder`** — partitions invariant boilerplate first / volatile per-company evidence last for provider prompt-cache hits (pure reorder preserving the content multiset); **(E) `routeModel`** — cheaper-first-pass-with-escalation **scaffold only** (NOT wired to live models; verification stages always strong; strong model at off; pending GATE5 per-measure/topic/language proof). Versioned kill-switch **`CONTEXT_COST_POLICY`** (`off` DEFAULT = current prod/byte-identical / `shadow` = savings telemetry, pack NOT modified / `on` = apply trim), `CONTEXT_COST_POLICY_VERSION=1`, `resolveContextCostPolicy()` fail-safe (unrecognized → off, total, never throws). Wiring: lever C gated at the A5 evidence seam in `analyzer.ts` (after `finalEvidence` settles) — at `off` the whole block is SKIPPED (no extra work, pack byte-identical); `shadow` logs savings without mutating; `on` applies; fail-open on error. Levers D/E module-only scaffolds (not wired). **Tests 11/11 (model-cost.test.ts): CC01 headline — every decisive pack passage ∈ keptIds + recall unchanged across negative/contradiction/qualifier/low-freq/table/single-decisive cases; CC02 only provably-redundant repeats trimmed; CC03 off-passthrough; CC04 shadow verdict-invariant; CC05 prefix multiset preserved + off byte-identical; CC06 routing (verify never cheap,**

escalation fires, off=strong); CC07 resolver fail-safe/frozen; CC08 determinism; CC-G3 byte-identical-at-off proof. tsc: 62 before → 62 after, **0 new** diagnostics in changed files. **At shipped-off default (`off`): reconstructed pack == original ⇒ Total cells Y→N = 0, N→Y = 0, Total Flips = 0** by construction. **GATE5 (2026-10-17, user-approved): default flipped `off→on` via code (`8725521`); env still overrides.** STEP 1 offline held-out proof over 1293's frozen corpus: **invariant holds 360/360, 0 violations** (decisive text never dropped), trim fired 326/360, aggregate pack-token reduction **25.39%** (3,920,993→2,925,442; harness `4156742`). Live corpus-replay batch 1294 off 1293 (canary-first co 9; snapshot gitSha=8725521 all 10 co ⇒ policy=on): 10/10, all ms=36. **Token/cost: prompt tok 13,937,501→12,221,478 (−12.31%), total tok −11.82%, cost $4.1684→$3.6851 (−11.59%), identical 1,580 calls — a REDUCTION (the B5 goal). Cell-flip vs 1293: Yes 67→70, Net +3, Y→N=4, N→Y=7, Total Flips=11** (consistency PASS; below noise floor A1 21/A2 20/A3 28/B1 19/A5 16; unstructured — 9 measures/5 co, 5 co zero flips; see 2026-10-17 section). **Total cells Y→N=4, N→Y=7.** | ✅ keep — now default **on** via `8725521` (GATE5 live batch 1294; Y→N=4, N→Y=7, Total Flips=11, Yes 67→70; −12.31% prompt tok / −11.59% cost; invariant 360/360; see 2026-10-17 section) | `CONTEXT_COST_POLICY=off` (or `=shadow`) restores the byte-identical pre-B5 pack with NO redeploy; or revert `8725521` (3-file default+comments+test, no schema change) then `154ff9b` to remove the whole B5 feature | | B6 | deec3a8 (code) · dynamic enabled 2026-10-18 | prod (all 4 svc, auto-deploy from main; production + sprint-10-preview, app+worker all SUCCESS on deec3a8; health `{"status":"ok","version":"3.0.0"}`). **ENABLED 2026-10-18 ~00:02-00:06 UTC: `CAPACITY_POLICY=dynamic` set on PRODUCTION app+worker via Railway GraphQL `variableUpsert` (user-approved). Env-only change — prod server code byte-identical to deec3a8 (`git diff deec3a8..f41d2af` = docs-only ledger commit; deployed HEAD f41d2af). Both prod svc redeployed SUCCESS on f41d2af; health `{"status":"ok","version":"3.0.0"}`; variable readback worker+app both `=dynamic`; worker startup log `[Worker] CAPACITY_POLICY=dynamic replicaCount=1 static=4 dynamic=4 effective=4 withinBudget=true maxSafeReplicas=40` ⇒ dynamic resolved == static == 4 at current replicaCount=1 (behaviourally identical now; only lowers at ≥5 replicas, never raises). Total cells Y→N=0, N→Y=0 by construction (worker concurrency change, scoring untouched, no paid batch).** | n/a — default `static` is a behavioural NO-OP (worker concurrency === prior prod constant `MAX_CONCURRENT`); no paid batch required to prove equivalence (code-path argument below) | WP14/B6 capacity experiments — connection-budget-aware worker concurrency (a disciplined capacity-TUNING method + ONE dynamic element; NOT a closed-loop autoscaler and NOT a replica-count controller — replica COUNT stays the user's manual Railway lever). **Bottleneck classified + every DB connection inventoried** (`gate_b6_out/b6_bottleneck_and_connection_inventory.{md,json}`): the app

runs on its OWN Railway Postgres, LIVE-confirmed `max_connections=100 − superuser_reserved_connections=3` ⇒ **97 usable** is the binding ceiling (this is the documented 88-vs-97 class of failure, NOT the Abacus 25/DB default). Prod pools: `db.ts/PG_POOL_MAX`=20 (app+worker) + 4 small gated pools (llm-usage/stage-profile/paid-call=2, run-manifest=1) ⇒ per-worker worst case **27**, per-app **26**; 2 app + 2 worker = **106 > 97** in the saturated worst case. New pure DB-free, no-company/topic module **`server/lib/capacity-config.ts`**: `effectiveConcurrency = clamp(floor((DB_CONNECTION_BUDGET − reservedOverhead) / replicaCount / connectionsPerJob), 1, HARD_MAX)` with in-code defaults budget=90, overhead=10, cpj=2, HARD_MAX=10 (=today's static concurrency ⇒ dynamic can only ever LOWER, never raise). **Ceiling invariant** `effectiveConcurrency × replicaCount × connectionsPerJob + reservedOverhead ≤ DB_CONNECTION_BUDGET` GUARANTEED by construction for replicaCount ∈ [1, maxSafeReplicas=40], exposed via `checkCeilingInvariant()` (beyond 40 it returns false — a documented "cut replicas" signal, never a silent breach). Versioned kill-switch **`CAPACITY_POLICY`** (`static` DEFAULT = current prod/byte-identical / `shadow` = compute+log budget-derived value, still USE static / `dynamic` = apply budget-derived value), `CAPACITY_POLICY_VERSION=1`, `resolveCapacityPolicy()` fail-safe (unrecognized → static, total, never throws). Wired ONLY at the worker-concurrency seam (`server/worker.ts` line 53/1570 — `concurrency: EFFECTIVE_CONCURRENCY`); at default `static` this resolves to exactly `MAX_CONCURRENT` ⇒ byte-identical. Does NOT touch analyzer/evidence/scoring and does NOT change any pool max size (approval-gated tuned-matrix decision). One-variable matrix harness `server/scripts/gate_b6_capacity_matrix.ts` prints safe operating point for any replica count. **Tests 13/13 (capacity-config.test.ts): CAP01 invariant holds across replicaCount 1..maxSafe × budget sweep + CAP01b infeasibility flagged beyond maxSafe + CAP01c known points (r=1→10,2→10,4→10,8→5,10→4,16→2,20→2,40→1); CAP02 default=static + static/shadow byte-identical; CAP03 dynamic only lowers ≤HARD_MAX; CAP04 env override; CAP05 fail-safe; CAP06 determinism; CAP07 purity/shadow-computes-not-applies.** Adjacent adaptive-concurrency suite 13/13 green. tsc: 62 before → 62 after, **0 new diagnostics in changed files**. **At shipped default (`static`): worker `concurrency` is literally the same `MAX_CONCURRENT` constant; analyzer/evidence/scoring surface untouched ⇒ Total cells Y→N = 0, N→Y = 0, Total Flips = 0 by construction** (no scoring surface mutated ⇒ no paid batch needed). **Total cells Y→N=0, N→Y=0.** | ✅ keep (default `static` = byte-identical no-op; `dynamic` is the user's approval-gated lever) | `CAPACITY_POLICY=static` is the default/no-op (set `=dynamic` to enable; no redeploy if env settable); or revert `deec3a8` (4-file self-contained module+test+harness+worker-seam wiring + gate_b6_out artifacts; no schema/pool/env change) | | A9 | 8aebe6b (code) | prod (all 4 svc, auto-deploy from main; production + sprint-10-preview, app+worker all SUCCESS on 8aebe6b; health `{"status":"ok","version":"3.0.0"}`; deployed

SHA == tested SHA on all 4 — full `8aebe6bb6dfbdbd26e7c563b8ee34af0b55fd13a`). | n/a — governance/reporting only, verdict-invariant; no paid batch (code-path argument below) | WP15/A9 truthful publication & release-evidence governance. New pure, DB-free, no-company/topic module **`server/lib/truthful-publication.ts`** (+ `truthful-publication.test.ts`). (a) **One canonical publication record accessor** `buildPublicationRecord()` — the single shape every reader surface (API/UI/CSV/Word/PDF) derives from, so verdict/evidence/research-status can never diverge (PU01, CV07). Links each assessment to framework version + source scope + as-of (`PublicationScope`), coverage/researchStatus (reuses `shared/research-state.ts` CV01–CV07 incl. `summarizeResearchStatus`/`researchStatusCsvColumns`), evidence IDs, and run provenance (reuses `server/lib/run-manifest.ts` `RunManifest`/`hashObject`/`canonicalize` — NOT a parallel system). (b) `assertTruthfulCompleteness()` — an incomplete assessment is NEVER rendered/exported as comprehensive. (c) **Immutable snapshots** `publishSnapshot()` (deep-frozen, hash-anchored) + **explicit reason-required revisions** `reviseSnapshot()` (links `revisionOf`+increment; original never mutated; throws on empty reason ⇒ no silent recompute) + `verifySnapshotIntegrity()`. (d) **Separate release record types** (candidate-review/approval/merge/deploy/validation/acceptance): `classifyCommitAsReleaseEvidence()` returns a HISTORICAL merge record (a commit message is never approval); `recordApproval()` throws if `fromCommitMessage` or missing approver/ref; `preserveFeatureBranchText()` keeps feature-branch-only text as dated history; `currentReleaseStatus()` ignores historical records. (e) **Release-gate evidence** `evaluateReleaseGate()` keeps the RAW non-deduped lists (tests+skips, TS diagnostic multiplicity, mutation checks, packaging) ALONGSIDE a dedup summary — relevant text never dropped; FAILS (never exit-0 skip) on test fail / unreasoned skip / diagnostics over baseline (raw count) / missing Chromium (PU04) / mutation blind spot (PU05) / missing worker asset (PU06). `reconcileRuntimeWithTested()` (PU03 — deployed SHA == tested SHA). (f) **WP23 measurement baseline** keyed by framework/topic/language/sector with `incompleteResearchRate` tracked SEPARATELY (no global-average hiding). Versioned kill-switch **`PUBLICATION_POLICY`** (`passthrough` DEFAULT = non-mutating, byte-identical to today / `annotate` = APPEND additive governance+research-status columns, existing fields untouched / `shadow` = also non-mutating reader surface), `PUBLICATION_POLICY_VERSION=1`, `resolvePublicationPolicy()` fail-safe (explicit > env > default; unrecognized/empty → `passthrough`, total, never throws). **Cross-surface routing:** the TS accessor is the single canonical verdict/status source; a committed Python/TS parity test pins `server/scripts/export_ai_csv_from_snapshot.py:norm_verdict` against TS `normalizeVerdict` cell-for-cell (CSV/Word/PDF can’t drift) — no invasive exporter rewrite (preserves byte-identical default). **Tests 18/18 (truthful-publication.test.ts): PU01 cross-surface consistency + PU01b**

incomplete≠comprehensive + PU01c Python/TS verdict parity; PU02 immutable snapshot + explicit revision; PU03 SHA reconcile; PU04/PU04b missing Chromium + unreasoned-skip fail; PU05 parent-cancellation mutation sensitivity; PU06 worker-asset packaging; TS-baseline gate; RAW-EVIDENCE preserved (non-deduped); release-record types (commit≠approval, feature-branch=history); KILL-SWITCH default passthrough byte-identical (5 base CSV cols, no governance block); NON-MUTATION (input never mutated, verdict uses only existing normalizeVerdict); RESOLVER fail-safe; ROUND-TRIP serialization; WP23 subgroup baseline. Legacy suites green (capacity-config 13/13, run-manifest 7/7, snapshot-durability 12/12). tsc: **62 before → 62 after, 0 new diagnostics in changed files**. **At shipped default (`passthrough`): no published-score surface is mutated — the verdict field is only the pre-existing canonical `normalizeVerdict` every surface already applies, and passthrough emits no new/changed reader field ⇒ Total cells Y→N = 0, N→Y = 0, Total Flips = 0 by construction** (reporting/governance only; no analyzer/evidence/scoring path touched ⇒ no paid batch needed). **Total cells Y→N=0, N→Y=0.** | ✅ keep (default `passthrough` = byte-identical no-op; `annotate` is the opt-in governance surface) | `PUBLICATION_POLICY=passthrough` is the default/no-op (set `=annotate` to surface governance columns; no redeploy if env settable); or revert `8aebe6b` (2-file self-contained module + test; no schema/pool/env/exporter change) | | B-dur | — | — | — | — | pending | — |

> Circuit-breaker: after 2 failed attempts on any single WP, mark BLOCKED and move on. Never auto-invoke quarantine-recovery override. Stop and surface before any Trial 2.

Comparison methodology — cell-level flips (why net Yes counts are insufficient)

Earlier result rows reported only the **number of Yes verdicts** per run (e.g. A2 58 vs baseline 56). A net Yes count hides **offsetting cell changes**: a run can flip several cells No→Yes and several others Yes→No, so a small net can sit on top of a large amount of actual churn — and a genuine regression (a measure that broke) can be masked by an unrelated improvement elsewhere. Net Yes alone therefore both understates churn and can hide direction-specific harm.

We compare two runs over the shared **company × measure** cell set (10 companies × 36 measures = 360 cells; identical set is asserted, not assumed) and report, per company and in aggregate:

- **Yes (baseline) / Yes (variant)** — Yes-verdict counts in each run.
- **Net** = Yes(variant) − Yes(baseline).

- **Y→N** = cells that were **Yes in baseline but No in variant** (direction: lost Yes).
- **N→Y** = cells that were **No in baseline but Yes in variant** (direction: gained Yes).
- **Total Flips** = Y→N + N→Y — the real count of cells whose verdict changed.

Internal consistency (always holds): **Net == N→Y − Y→N**. "No" here means *any* verdict that is not exactly `Yes` (No / Partial / Insufficient evidence / Scoring error).

Numbers below are produced by `server/scripts/compare_run_flips.ts` reading the immutable `analysis_results` batch snapshots (read-only); they are reproducible with `node --import tsx server/scripts/compare_run_flips.ts <baseline> <variant>`.

A2 (batch 1286) vs WP01 baseline (batch 1284) — corrected table

Cell set: 10 companies × 36 measures = 360 cells. Source: `analysis_results` snapshots.

Company ID	Company	Yes (baseline)	Yes (variant)	Net	Y→N	N→Y	Total Flips
9	CSX Corporation	3	2	−1	1	0	1
10	MAGNA INTERNATIONAL INC.	10	12	+2	1	3	4
11	THE SOUTHERN COMPANY	2	5	+3	0	3	3
12	S&P GLOBAL INC.	14	14	+0	3	3	6
13	MARTIN MARIETTA MATER	0	0	+0	0	0	0

Company ID	Company	Yes (baseline)	Yes (variant)	Net	Y→N	N→Y	Total Flips
	IALS, INC.						
14	DARDEN RESTAURANTS, INC.	0	0	+0	0	0	0
15	BCE INC.	14	14	+0	1	1	2
16	STRYKER CORPORATION	7	8	+1	0	1	1
17	AMPHENOL CORPORATION	0	0	+0	0	0	0
18	AFLAC INCORPORATED	6	3	−3	3	0	3
ALL	**Aggregate (10 companies)**	**56**	**58**	**+2**	**9**	**11**	**20**

Consistency check (Net == N→Y − Y→N): **PASS** (+2 == 11 − 9).

Reading: the +2 net Yes is the surface of **20 changed cells** (9 lost, 11 gained). The churn concentrates in S&P Global (6 flips, net 0), Magna (4), Southern (3) and Aflac (3 flips, all Yes→N). The three all-No companies (Martin Marietta, Darden, Amphenol) show **zero** flips — baseline-stable, confirming no over-rejection there.

For reference, **A1 (batch 1285) vs baseline (1284)**: Yes 56→55, Net −1, Y→N=11, N→Y=10, **Total Flips=21** (consistency: −1 == 10 − 11, PASS). And **A2 vs A1 (1286 vs 1285)**: Net +3, Y→N=7, N→Y=10, Total Flips=17 — the 7 Y→N here reproduce exactly the earlier `a2_cmp.json` `y2nList`, which had lacked the N→Y side and totals.

A3 (batch 1287) vs WP01 baseline (batch 1284)

Cell set: 10 companies × 36 measures = 360 cells. Source: `analysis_results` snapshots.

Company ID	Company	Yes (baseline)	Yes (variant)	Net	Y→N	N→Y	Total Flips
9	CSX Corporation	3	5	+2	0	2	2
10	MAGNA INTERNATIONAL INC.	10	12	+2	1	3	4
11	THE SOUTHERN COMPANY	2	3	+1	1	2	3
12	S&P GLOBAL INC.	14	13	−1	4	3	7
13	MARTIN MARIETTA MATERIALS, INC.	0	0	+0	0	0	0
14	DARDEN RESTAURANTS, INC.	0	5	+5	0	5	5
15	BCE INC.	14	16	+2	1	3	4
16	STRYKER CORPORATION	7	7	+0	0	0	0

Company ID	Company	Yes (baseline)	Yes (variant)	Net	Y→N	N→Y	Total Flips
17	AMPHENOL CORPORATION	0	0	+0	0	0	0
18	AFLAC INCORPORATED	6	3	−3	3	0	3
ALL	**Aggregate (10 companies)**	**56**	**64**	**+8**	**10**	**18**	**28**

Consistency check (Net == N→Y − Y→N): **PASS** (+8 == 18 − 10).

A3 (batch 1287) vs A2 (batch 1286)

Company ID	Company	Yes (A2 1286)	Yes (A3 1287)	Net	Y→N	N→Y	Total Flips
9	CSX Corporation	2	5	+3	0	3	3
10	MAGNA INTERNATIONAL INC.	12	12	+0	1	1	2
11	THE SOUTHERN COMPANY	5	3	−2	2	0	2
12	S&P GLOBAL INC.	14	13	−1	4	3	7
13	MARTIN MARIETTA	0	0	+0	0	0	0

Company ID	Company	Yes (A2 1286)	Yes (A3 1287)	Net	Y→N	N→Y	Total Flips
	MATERIALS, INC.						
14	DARDEN RESTAURANTS, INC.	0	5	+5	0	5	5
15	BCE INC.	14	16	+2	1	3	4
16	STRYKER CORPORATION	8	7	−1	1	0	1
17	AMPHENOL CORPORATION	0	0	+0	0	0	0
18	AFLAC INCORPORATED	3	3	+0	0	0	0
ALL	**Aggregate (10 companies)**	**58**	**64**	**+6**	**9**	**15**	**24**

Consistency check (Net == N→Y − Y→N): **PASS** (+6 == 15 − 9).

Reading (A3 is shadow-mode — these flips are run variance, not A3 behaviour). A3 deployed with `ELIGIBILITY_ENFORCE` **unset** in prod, so the composition logic attaches an eligibility decision to each `MeasureResult` but mutates **no verdict**; by construction A3 cannot have produced any of the 28/24 flips above — they are the same run-to-run model/corpus variance seen between every pair of runs (cf. A1-vs-base 21 flips, A2-vs-base 20 flips). Direction confirms **no collapse or over-rejection**: the net is a Yes **increase** (+8 / +6), not a loss; the three structurally all-No cos (Martin Marietta, Darden, Amphenol) show zero regression; **Darden 0→5** is a gain, not a collapse; and the single loss — **Aflac −3 vs baseline** — is not new to

A3: Aflac dropped 6→3 under A2 as well (see A2 table above) and is **stable A2→A3 (3→3, zero flips)**, i.e. a reproducible corpus artefact independent of this change. The STEP-5 stop-rule (halt on measure-level collapse / over-rejection / regression) is therefore **not triggered**; no second batch was run.

Correctness is established by the unit suite, not the live run. Because the live batch is shadow-mode, the EP01–EP06 guarantees (rejected third-party stays rejected through every path; a subsidiary can't support a parent-wide claim unless scope permits; a current-year question doesn't inherit a superseded report; issuer-filed vs analyst-commentary is distinguished; cache reuse can't carry stale eligibility; incomplete provenance → repair/review, never authority assumption) are proven by the 20 passing tests in `server/lib/eligibility-decision-a3.test.ts`, reproducible with `DATABASE_URL=postgres://stub:stub@127.0.0.1:1/stub npx tsx server/lib/eligibility-decision-a3.test.ts`.

> **GATE3 caveat (no held-out human labels).** There is no adjudicated human-labelled set for this roster, so A3 acceptance rests on (a) the EP01–EP06 unit tests encoding the composition guarantees and (b) demonstrably safe, baseline-stable live behaviour in shadow mode — **not** on measured agreement with human ground truth. Enabling enforcement (`ELIGIBILITY_ENFORCE=on`) should be gated on such a held-out check.
>
> **Carried-forward constraint (recovery-boundary guard + small-PDF runtime gap).** This open constraint belongs to **A4/WP05** (subprocess PDF extraction — EX04/EX06). A3 touches none of the extraction path (`pipeline.ts` / `processor.ts` are untouched; A3 adds a standalone module, one optional `MeasureResult` field, and an env-gated shadow block in `analyzer.ts`), so it does **not intersect** A3 and is **carried forward to A4** unchanged.

Step-1 trace findings (A3 composition gap, confirmed before building)

The plan flagged wrong-entity reinsertion as a *carried-forward* concern to re-verify. Trace confirmed the app already has the individual controls — `classifyProvenance` (`server/lib/provenance.ts`) labels issuer-self / issuer-filed / third-party sources, and `issuer-resolver.ts` derives entity aliases — but there was **no single immutable decision composed from them and carried through scoring**: provenance was classified ad hoc at gate time and not re-pinned per target, so a reused passage or a similarly-named / subsidiary / superseded source had no end-to-end object asserting eligibility. A3 fills exactly that composition gap (compose once → freeze → recompute per target → bounded repair that can't override exclusion), which is why it is a *composition* guardrail over existing controls rather than a replacement for them.

Baseline run (WP01 reference) — ESTABLISHED

- **Batch id:** 1284
- **Deployed commit (all 4 services):** 328d025 (= measurement-foundations code 2b91d86 + docs-only commits; carries stage_profile_events / paid_call_events / run_manifests telemetry sinks)
- **Roster:** frozen 9–18 (CSX, Magna, Southern Co, S&P Global, Martin Marietta, Darden, BCE, Stryker, Amphenol, Aflac)
- **Workspace 3, framework 18 (36 measures), provenance list_id=7**
- **Enqueue:** production-canonical POST /api/admin/resume-analysis (createBatchRun→createAnalysisJobs→addBatchJobs), 10/10 enqueued
- **Pre-run state:** CLEAN SLATE — documents + measure_scores purged, status reset to idle for all 10 (prior state snapshotted to /home/ubuntu/baseline_prerun_snapshot.json: 180 measure_scores, 309 docs). Clean-slate chosen so every company runs full discovery→fetch→extraction→retrieval→scoring (uniform/reproducible starting state + complete stage telemetry + maximal regression coverage); harness-only reset, pipeline/scoring code byte-identical to production.
- **Sole queue consumer:** prod worker (3 replicas); dev worker off shared “analysis” queue (sleep infinity, numReplicas 1).
- **Enqueued at:** 2026-10-15 (resume-analysis HTTP 200)
- **Pass bar (from batch-1275):** 10/10 first-attempt completions, each ≤35 min from claim; any retry voids pass; no accuracy/completeness regression.
- **Status:** ✅ COMPLETE / CAPTURED — 2026-10-15T13:29:51Z.

Baseline captured results (batch 1284) — this is the WP01 reference for all comparisons

- **Completion:** 10/10 completed, 0 failed, **all first-attempt (attempts=1)**. Wall-clock **46m16s** (8-concurrency, 1 prod worker × 3 replicas).
- **Per-company claim→complete:** min 12m29s (Aflac), max 26m42s (Martin Marietta). **NONE exceeded 35 min.** ✅ PASS BAR MET.
  - CSX 23m15s · Magna 20m18s · Southern 20m28s · S&P 26m17s · Martin Marietta 26m42s · Darden 24m43s · BCE 24m05s · Stryker 19m33s · Amphenol 26m16s · Aflac 12m29s.
- **Scores (deterministic reference — keep byte-stable for equivalence WPs):**

id	company	score	met/36	ms
9	CSX	8	3	36
10	Magna	28	10	36
11	Southern Co	6	2	36
12	S&P Global	39	14	36
13	Martin Marietta	0	0	36
14	Darden	0	0	36
15	BCE	39	14	36
16	Stryker	19	7	36
17	Amphenol	0	0	36
18	Aflac	17	6	36

Aggregate: sum_score=156, sum_met=56, all companies ms=36 (full coverage). Verdict distribution over 360 measure_scores: **No=304, Yes=56**.

- **Cost / telemetry:** total paid LLM cost **$3.9476** (deepseek 1825 calls $3.9464 + openrouter 1 escalation $0.0012); 13,704,064 tokens; 1826 LLM calls, 61 browser (evomi_proxy), 2161 search (serper). 104 stage_profile_events. run_manifest run_id=batch-1284, mode=fresh, hash=fe36c21278c271c6.
- **Interpretation:** three companies scored 0/0 (Martin Marietta, Darden, Amphenol) — all completed with ms=36 (every measure adjudicated), i.e. legitimate "all-No" results, not failures. This all-No cohort is exactly where WP02/A1 research-state accounting (distinguish "looked & found nothing" from "never got doc") will add signal without changing verdicts.

WP04 / A3 — GATE3 label set (full supporting-quote-set eligibility) — ESTABLISHED

What changed. Commit `9e94507` ("feat(WP04/A3): full supporting-quote-set eligibility basis + quote-correction-primary enforcement") is on `origin/main` and auto-deployed to all four Railway services (shadow: `ELIGIBILITY_ENFORCE` unset). A3's eligibility basis now equals the **verdict's basis** — it evaluates the **entire supporting-quote set** of each positive cell (new pure/total `composeCellEligibility` in `server/lib/eligibility-decision.ts`), not just the single cited quote. Remedy order is **quote-correction primary, verdict-change last resort**: if any supporting quote

is eligible, promote it (`A3-CORRECT`, verdict unchanged); only if **all** quotes are hard-excluded does the verdict downgrade to No/0 (`A3-ENFORCE DOWNGRADE`).

GATE3 forecast — method (replaces the "no held-out labels" caveat with a measured forecast). Rather than a human-labelled set (none exists for this roster), the label set replays the exact enforcement composer over **real production evidence**: batch **1287** snapshot (`analysis_results` id 803, roster 9–18, policy v2). Persisted snapshots drop quote `sourceUrl`, so URLs were **recovered** by matching each quote's `{text, source}` back to the batch corpus (`batch_corpus→documents/document_content`) via NFKC-normalised anchor-window containment. **Recovery rate 99.34% (150/151 supporting quotes; 1 unresolved)** — high enough that the offline replay is rigorous, so **no paid batch was spent** (the fail-silent "URL-drop ⇒ REVIEW" artifact that made the first offline attempt unusable is fully eliminated: **0 cells excluded as artifacts**).

Label buckets (64 positive cells, all currently Yes):

Bucket	Cells	Share	Meaning
STAY	56	87.5%	cited quote already eligible — no change
QUOTE_CORRECT	5	7.8%	eligible supporting quote promoted; **verdict unchanged**
VERDICT_CHANGE	3	4.7%	all quotes hard-excluded → downgrade (last resort)
REVIEW (genuine)	0	0.0%	none eligible, some unknown-provenance
excluded (URL-drop artifact)	0	n/a	—

Directional verdict flips implied by enforcement (per standing reporting constraint):

Direction	Cells	Of which genuine	Of which provenance-attribution artifact
Yes→No	**3**	**0**	**3**

Direction	Cells	Of which genuine	Of which provenance-attribution artifact
No→Yes	**0**	0	0
Total cells changing verdict	**3**	**0**	**3**

(No→Yes is structurally 0: A3 only acts on positive cells.) Per-company detail and the per-quote exclusion-kind table are in `/home/ubuntu/gate3_labels.md` / `.json`; all 3 Yes→No flips are **S&P Global (co 12)**.

User expectation — CONFIRMED. The governing expectation was that making A3 evaluate the full quote set *"should almost always result in the quote being changed … rather than the verdict being changed."* Among the 8 actionable cells, **quote-correction (5) dominates verdict-change (3)**; VERDICT_CHANGE is 4.7% of classified cells (rare). The data confirms the expectation.

Root-cause of the 3 VERDICT_CHANGE cells — provenance-attribution gaps, not genuine ineligibility. Every excluded quote in the 3 flips is **S&P Global's own disclosure** (2025/2026 Proxy Statements) whose recovered URL the classifier cannot map to the issuer: unmapped **Q4 Inc IR-distribution CDN** tenants (`s29.q4cdn.com`, kind `ir_cdn`) and **search-engine redirect masks** (`www.google.com/goto`, kind `redirect_mask`). These are issuer-self sources mis-scored as third-party — i.e. a classifier provenance gap, **not** evidence the verdict should flip. **Genuine verdict-change after removing the artifacts = 0.**

Enforcement enablement — GATED (two prerequisites, decision required)

Enabling `ELIGIBILITY_ENFORCE=on` **as-is** would apply 3 **false** Yes→No downgrades to S&P Global. That contradicts the data (0 genuine ineligibility) and the project goal ("legitimate results per measure, not fail-silent downgrades"). Enforcement is therefore gated on:

1. **Provenance-gap closure (recommended first; code, not env).** Map the issuer's Q4/IR-CDN tenants into `related_domains` and resolve `www.google.com/goto` redirect masks to their target before provenance classification, so these 3 issuer disclosures score as issuer-self. After this, the forecast becomes **STAY 56 / QUOTE_CORRECT 5 / VERDICT_CHANGE 0** and enforcement produces **zero** verdict flips on this roster — pure quote-correction, exactly the intended

behaviour. *(Distinct scope from the approved A3 work; awaiting go-ahead.)*

2. **Production write mechanism.** Flipping `ELIGIBILITY_ENFORCE=on` in prod needs Railway env-var write access (current CLI token here is unauthenticated), **or** approval to change the in-code default to enforce-on (ships via `git push` to main, which is how `9e94507` already deployed; kill switch `ELIGIBILITY_ENFORCE=off` + `ELIGIBILITY_POLICY=v1` remain). Code deploys work from here; **env-var writes do not.**

Carried forward unchanged: WP05/A4 (behavioral recovery-boundary guard + small-PDF runtime gap) remains OUTSTANDING and does not intersect A3.

WP04 / A3 — Framework-level provenance fix SHIPPED + enforcement ENABLED + validated — ESTABLISHED (2026-10-16)

Decision taken. The provenance-gap closure of the section above was executed, and the in-code enforcement default was flipped to **enforce-ON** (both prerequisites resolved). Crucially, the fix was made **framework-level, with zero per-company / per-topic data** — the earlier proposed 1-row `related_domains` tenant declaration for S&P Global (id 12) was **REJECTED by the user as company-specific** and was **never written** (production DB guard also blocked it). It is fully superseded.

Commits on `origin/main` (auto-deploy to all 4 Railway services; dev==prod): `58236a2` (GATE3 label-set docs) → `e06a78f` (enforcement default-ON + redirect-mask unwrap + IR-platform recognition) → **`15bbcc0`** ("feat(provenance+A3): framework-level full-name phrase identity + IR-tenant propagation through eligibility composer") — the superseding framework fix.

What `15bbcc0` ships (all name-derived, company-/topic-agnostic, NO data rows): 1. **Full-name phrase identity** (`server/lib/provenance.ts` `fullNamePhraseMatch`) — an issuer's complete legal name minus generic corporate-form suffixes (Inc/Corp/PLC/Co) is matched as a contiguous phrase against a document's title/URL-path/content, so an issuer with no distinctive single token and no ticker in the URL (e.g. S&P Global, ticker SPGI, on `s29.q4cdn.com/690959130/.../S-P-Global-Inc-2026-Proxy-Statement.pdf`) self-confirms its own IR document. Guarded ≥2 tokens & ≥6 joined chars so short names (AT&T, Visa) do not over-fire. 2. **IR-tenant propagation through the A3 eligibility composer** (`composeCellEligibility`, two-pass, order-independent) — the existing R5c `knownIrTenants` cache (write-through whenever ANY source self-confirms a

Q4/IR-CDN tenant) is now threaded through the composer and shared across all of a company's sources (one cache per company in `analyzer.ts` AND in the GATE3 builder, so offline == live). A confirmed tenant then covers sibling documents on the same tenant that carry no self-identifying name. Cross-company reuse guard (`companyMatchesBinding`) still blocks a different issuer borrowing another's bound tenant. 3. **Enforcement default-ON** (`server/lib/analyzer.ts`) — unset/empty `ELIGIBILITY_ENFORCE` now enforces; kill switches intact: `ELIGIBILITY_ENFORCE=off` restores shadow mode, `ELIGIBILITY_POLICY=v1` restores legacy passthrough.

Tests: `eligibility-decision-a3.test.ts` (34) + `provenance.test.ts` (49) pass, incl. 6 new R5c cases (order-independent sibling propagation; cross-company reuse over-fire guard), plus kill-switch and default-ON cases.

Validation — deterministic isolation over intact snapshot 1287 (free, zero-confound)

The fix lives **purely in post-scoring eligibility composition** — it changes neither retrieval nor LLM scoring. The defensible validation is therefore a **deterministic replay of the fixed composer over a fixed snapshot**, which isolates the fix's directional flips with **zero LLM-noise confound**. Snapshot **1287** (analysis_results id 803) is the frozen roster (FW18, companies 9–18, full run) with a **byte-intact corpus** (457/457 docs join, 59.19M chars recovered at 99.34% URL-recovery).

Bucket (64 positive cells, snapshot 1287)	Pre-framework-fix	**Post-fix (`15bbcc0`, enforce-ON)**
STAY	56	**63**
QUOTE_CORRECT	5	**1**
VERDICT_CHANGE	3	**0**
REVIEW (genuine)	0	**0**

Directional verdict flips from enabling enforcement (standing reporting constraint): - **Yes→No = 0** (pre-fix would have been **3**, all S&P provenance-attribution artifacts, **0 genuine**) - **No→Yes = 0** (A3 acts only on positive cells) - **Total cells changing verdict = 0**

Only S&P Global (id 12) changed between pre-fix and post-fix; the other 9 companies are **byte-identical — zero regressions**. The one QUOTE_CORRECT (Magna id 10) is the intended remedy (promote an eligible quote, verdict unchanged). Evidence: `deliverables/release-evidence/provenance-gap-closure/AFTER_framework-phrase-and-propagation.{json,md}`.

Why NOT a fresh paid batch — the LLM noise floor swamps the signal

Two fresh runs of the **identical frozen roster with no code change** — baseline **1284** (snapshot 800) vs **1287** (snapshot 803), both run 2026-10-15 — differ by **28 cell flips**:

	Yes (1284)	Yes (1287)	Net	Y→N	N→Y	Total Flips
Aggregate (10 cos)	56	64	+8	**10**	**18**	**28**

This 28-flip run-to-run variance is **pure LLM nondeterminism** (no code differs between the two runs). It is ~10× the fix's isolated effect (0–3 cells), so a fresh paid batch would make the fix's directional flips **unattributable** — the noise would mask the signal. This is why the deterministic composer replay over a fixed snapshot is strictly more defensible for an eligibility-layer change, consistent with the isolation-reliability discipline.

Why baseline 1284 itself could not host the deterministic replay

Baseline **1284**'s underlying corpus **documents have been pruned** from the DB: its `batch_corpus` id-rows survive (333 across 10 companies) but the `batch_corpus → documents → document_content` join returns **0/333** (vs 457/457 intact for 1287). Free URL recovery over 1284 is therefore impossible (measured 0/140 quotes resolved). Snapshot 1287 — the **same frozen roster** — is byte-intact and is the correct validation snapshot. The raw-verdict snapshots for both batches survive (used in the noise-floor table above); only the corpus text for 1284 is gone.

Status: A3 COMPLETE. Framework-level provenance fix shipped (`15bbcc0`), enforcement enabled (default-ON, kill switches intact), validated at VERDICT_CHANGE=0 / Y→N=0 / N→Y=0 with zero regressions. **Next: WP05/A4** (behavioral recovery-boundary guard + small-PDF subprocess-extraction runtime gap, EX04/EX06).

A4 / WP05 — ESTABLISHED (2026-10-16): extraction / table / OCR / translation fidelity

Scope. Framework-level only — ZERO per-company/per-topic logic or data patches. All change confined to the extraction layer in `server/lib/processor.ts` plus a new test file `server/lib/extraction-fidelity.test.ts`. No schema change, no env-var change, no scoring/analyzer touched. Verdict-invariant by construction.

Files changed - `server/lib/processor.ts` — extraction-engine layer reworked (details below). - `server/lib/extraction-fidelity.test.ts` — new, EX01–EX07 + boilerplate-safety (10 tests). - `docs/implementation-run-ledger.md` — this section + A4 row.

Small-PDF runtime gap — ROOT CAUSE and framework fix. Empirically confirmed: `pdf-parse@1.1.4` bundles an old pdfjs whose xref reader REJECTS minimal hand-built xref tables ("bad XRef entry") — even after the worker's retry-once. A 767-byte fixture fails pdf-parse yet `pdftotext` recovers its full text. The earlier REJECTED approach was a size-special-case bound. The shipped fix is NOT size-based: a pure, exported outcome-classifier `classifyExtractionCompleteness({textLen,minChars,timedOut,failed,pagesAvailable?,pagesProcessed?})` keyed on the *outcome* (parser rejected structure vs. document genuinely empty vs. timed-out partial), not on byte size. "Parser rejected" → `unknown` (retry/review, never a false `empty`); clean run with no text → `empty`; timed-out with partial text → `partial`, never promoted to `complete`. The orchestrator `extractPdfWithFidelity` chains pdf-parse → pdftotext → adaptive OCR and feeds the classifier, so a small PDF rejected by pdf-parse is recovered by pdftotext and labelled `complete`.

Recovery-boundary guard — preserved by non-interference. The behavioural guard lives in `pipeline.ts` (`runCompanyPdfRecovery`, `remainingFetchBudgetMs`, `RECOVERY_WINDOW_FLOOR_MS`) and is proven by `company-pdf-recovery-boundary.test.ts` (B1-1..B1-7). `pipeline.ts` was NOT touched. Confirmed green: **B1-1..B1-7 all pass** post-change.

Subprocess timeout fidelity (EX06). Previously `extractViaPdftotext` silently returned partial stdout on timeout (accepted as success). Now `extractViaPdftotextDetailed` returns `{text, timedOut:true, failed:true}` on timeout; the classifier never promotes timed-out partial stdout to `complete`. `runProcessCollect` still captures partial stdout and flags `timedOut` (verified) — the fix is in how the outcome is *classified*, not in discarding the bytes.

Adaptive OCR (EX02, Lane-B speed gated on Lane-A fidelity). `extractViaOcrDetailed` is per-page: tries the `pdftotext -f p -l p` text layer first and skips OCR for any page already carrying ≥20 chars (text pages preserved verbatim), else rasterises just that page (`pdftoppm`) and OCRs it (`tesseract`). No 8-page cap — all pages processed, budget-bounded; flags `partial` when budget/scope cuts it short (`pagesProcessed < pagesAvailable`). Strictly ⊇ old content.

Table / translation / boilerplate fidelity (f). Asserted against existing framework helpers (not rewritten): `normaliseTableHorizonMarkers` annotates bullet-only horizon rows with their column relationship; `translation.ts` keeps the aligned derivative (`looksNonEnglish` detects foreign text; translation never replaces the original source carrying

numbers/negation/scope); `stripRepeatedStructuralBlocks` dedupes repeated headers but NEVER strips a repeated line containing a quantified figure (policy/decisive note preserved).

Test results (EX01–EX07 + safety) — all green, 10/10 | Test | Result | Proves | |——|———|———-| | EX01 | ok | decisive evidence on a later page survives (no truncation) | | EX02 | ok | mixed doc: text page verbatim AND scanned page recovered by adaptive OCR | | EX03 | ok | table units/headers preserved; horizon row→column relationship annotated | | EX04 (small) | ok | pathological pdf-parse-rejected xref recovered via pdftotext, labelled `complete` (no false empty) | | EX04 (large) | ok | benign 40-page PDF fully extracted, first AND last page markers present (no silent truncation) | | EX05 | ok | translation is an aligned derivative; detector flags foreign text, original source preserved | | EX06 (capture) | ok | `runProcessCollect` captures partial stdout on timeout and flags `timedOut` | | EX06 (rule) | ok | completeness rule NEVER promotes timed-out partial stdout to `complete` | | EX07 | ok | downstream passage recall = 10/10 = 100% across ordinary/multipage/table/OCR/non-Latin | | A4 safety | ok | boilerplate dedupe removes repeated headers but KEEPS a repeated policy line with a figure |

Regression suites — 12/12 green: `company-pdf-recovery-boundary` (B1-1..B1-7, guard preserved), `pdf-parse-subprocess-equivalence` (B6 byte-identical + B3/B2 slot handling), `stalled-extraction-boundary`, `table-horizon-normaliser`.

Typecheck: `tsc --noEmit` — 0 new diagnostics in `processor.ts` / `translation.ts` / `extraction-fidelity.test.ts` (repo has unrelated pre-existing errors in other files).

Cell-flip impact: n/a — no live batch run. The change is verdict-invariant by construction: `extractTextFromPdf` is a thin wrapper returning `ExtractionFidelity.text` byte-for-byte to all 11 existing callers; the fidelity record is additive metadata consumed by nothing in the scoring path. **Total cells changing Y→N = 0, N→Y = 0** (no scoring surface touched).

Boundary noted (not a gap): the `ExtractionFidelity` record (completeness/pages/engine/language/ truncation) is produced and available but not yet threaded into `storage.recordFetchSuccess` / `extraction_state` — intentionally out of scope for this surgical, verdict-invariant change. The `extraction_state` column already supports `partial`; wiring the completeness label through is a follow-up and would remain verdict-invariant.

A5 / WP06 — ESTABLISHED (2026-10-16): retrieval & evidence-pack recall

Scope. Framework-level only — ZERO per-company/per-topic logic or data patches. A valid document can sit in the corpus while its DECISIVE passage never reaches the model prompt; smarter models cannot fix missing input. A5 does NOT add a new retriever — it (a) MEASURES passage-level recall of decisive passages AT THE ACTUAL PROMPT BUDGET, and (b) closes measured misses with a BOUNDED, append-only evidence-expansion that EXTENDS the existing BM25/deep-read path and COMPOSES with A3 eligibility and A4 fidelity — never by shrinking top-k or pack length.

Files changed - `server/lib/evidence-recall.ts` — new. Pure, DB-free, LLM-free toolkit (oracle + recall + planner). - `server/lib/evidence-recall.test.ts` — new. RT01–RT06 + kill-switch-default + no-op-append (8 tests). - `server/scripts/evidence_recall_benchmark.ts` — new. Free/offline/deterministic recall benchmark. - `server/lib/analyzer.ts` — strictly gated wiring at the pre-scoring evidence seam (details below). - `docs/implementation-run-ledger.md` — this section + A5 row.

Kill-switch and shipped default. `RECALL_EXPANSION_POLICY` (resolved by `resolveRecallExpansionPolicy()`), three values: - **`off` — DEFAULT = current production behaviour.** The analyzer block is skipped ENTIRELY: no extra compute, no extra retrieval, the evidence pack is byte-identical, the verdict surface is unchanged. - `shadow` — compute recall telemetry + the expansion plan but DO NOT modify the pack (verdict-invariant; observability only). - `on` — APPLY the bounded expansion (behaviour-changing; append-only). Any unrecognised value falls back to `off` (fail-safe — a typo can never silently enable expansion). Verdict-changing retrieval is NEVER the default and is NOT set in the prod env. **To enable: `RECALL_EXPANSION_POLICY=on`** (or `=shadow` for telemetry); unsetting it restores current behaviour with no redeploy.

Decisive-passage oracle (framework-level). `computeDecisivePassages({measure,chunks})` marks a passage decisive when it carries ≥ 2 distinct decisive tokens OR a verbatim multi-word `evidenceKeyword` phrase. The decisive signal is drawn ONLY from the measure's own framework-authored fields (`title` + `evidenceKeywords`) — no company names, no topic hardcoding. Ranked (phraseHit, distinctTokens, stable id) and capped (`RECALL_DECISIVE_MAX`, default 12), so it is deterministic and bounded. This single definition is BOTH the recall ground-truth AND the candidate universe for expansion, so "what should have reached the model" and "what expansion may add" can never diverge.

Recall measurement at the actual prompt budget. `measurePackRecall({decisive,packPassageIds})` reports |decisive ∩ pack| / |decisive| by STABLE passage identity (`stablePassageId` = `docUrl#seq#sha1(text)`, mirrors the evidence-pack fingerprint scheme), plus

the exact misses in oracle order. `packPassageIds` are the ids of the chunks that ACTUALLY reached the pack shown to the scoring role (membership recovered by verbatim substring against the assembled pack text).

Bounded expansion planner (A3-composing, append-only, deterministic). `planBoundedExpansion(...)` fires only on DEMONSTRATED insufficiency (an eligible decisive passage exists in the corpus but missed the pack). It: - COMPOSES with A3: an injected eligibility predicate filters excluded material BEFORE selection, so expansion can NEVER reintroduce what A3 excluded (RT04). The analyzer injects the real A3 predicate (`composeEligibilityDecision` per candidate; `status==="excluded"` ⇒ ineligible), fail-open on error. - is BOUNDED: ≤ `RECALL_EXPANSION_MAX_PASSAGES` (default 6) and ≤ `RECALL_EXPANSION_MAX_CHARS` (default 12000) per measure — no unlimited retrieval. - NEVER shrinks top-k or pack length (append-only; the original pack is intact, order preserved). - dedups ONLY when a candidate adds no new decisive token AND its source link is already represented — so every unique decisive fact AND source link stays recoverable; qualifiers/negation in a distinct passage are never dropped (RT06). - is DETERMINISTIC: stable identity + stable ordering, re-runs byte-identical (RT05).

analyzer.ts wiring (strictly gated). At the pre-scoring seam (`finalEvidence = evidenceText`), guarded by `if (recallPolicy !== "off" && settings.useBm25Retrieval && evidenceText)`. When `off` the whole block is skipped (no `buildDeepIndex()`, no oracle, no plan) ⇒ zero added cost and byte-identical packs. When `shadow` it logs recall telemetry only; when `on` it appends `formatExpansionAppend(plan)` (a provenance- headed block) to `finalEvidence`. No existing pack content is removed or reordered.

Deterministic test results (RT01–RT06 + extras) — 8/8 pass | Test | Result | Proves | |——|———–|———–| | RT01 | ok | synonym/sector-language passage is decisive and recalled at the budget | | RT02 | ok | a low-frequency phrase-hit (table cell) is decisive and recoverable by bounded expansion | | RT03 | ok | negative/contradictory evidence is decisive and never suppressed | | RT04 | ok | bounded expansion composes with A3 — excluded material is never reintroduced | | RT05 | ok | stable passage identity and deterministic ordering under input shuffling | | RT06 | ok | dedup drops only pure duplicates; unique facts/links/negation preserved | | kill-switch | ok | default is `off` (current production behaviour, verdict-invariant) | | no-op append | ok | `formatExpansionAppend` returns `""` for a non-firing plan (byte-identical pack) |

Run: `DATABASE_URL=postgres://stub:stub@127.0.0.1:1/stub npx tsx server/lib/evidence-recall.test.ts` (the tests never touch the DB; the stub only satisfies the pool constructor).

GATE2 — free/offline/deterministic recall benchmark. `server/scripts/evidence_recall_benchmark.ts` builds the EXACT pack the

scorer sees using the REAL, deterministic `summarizeDocuments` (retrievalV2 BM25 corpus selection — NO LLM) + the REAL `buildEvidencePacksForCategory`, over the frozen fixture corpus (`var_corpus/*`), and reports passage-level recall + shadow recoverability. Read-only w.r.t. the production DB (needs only an empty throwaway `summary_cache`; a cache miss is the normal path). Full 13-company fixture roster, all 29 measures, 0 company errors:

Metric	Value
Companies evaluated	13 (ids 1, 20, 120, 22, 107, 21, 14, 18, 11, 12, 13, 102, 16)
(company × measure) cells	377
Decisive passages (corpus total)	4493
Recalled into the budgeted pack	3334
Micro-averaged recall	**74.2%**
Macro-averaged recall	74.3%
Decisive misses	1159
Recoverable by A5 bounded expansion (shadow)	537 (46.3% of misses)
Budget	corpusMaxChars 560000, top-k 45, evidenceMaxChars 45000
Resolved policy during run	`off` (expansion run in SHADOW for measurement; pack never modified)

Outputs: `var_out/evidence_recall_benchmark.{json,md}` (per-cell + aggregate). Re-runs are byte-identical.

Typecheck: `tsc --noEmit` — 0 new diagnostics in `evidence-recall.ts` / `evidence-recall.test.ts` / `evidence_recall_benchmark.ts` / `analyzer.ts` (repo has 62 unrelated pre-existing errors in other scripts/files).

Adjacent regression suites — all green: evidence-gate 14/14, evidence-gate-a2 17/17, eligibility-decision-a3 34/34, passage-retrieval 35/35, retrieval-stability 5/5, retrieval-review-queue (green). No regression in the retrieval/evidence-gate/eligibility surfaces A5 sits next to.

Cell-flip impact: n/a — no live batch run, and none is required. The shipped default `RECALL_EXPANSION_POLICY=off` skips the analyzer block entirely, so the evidence pack and verdict surface are byte-identical to current production. **Total cells changing Y→N = 0, N→Y = 0** (no scoring surface touched at the default). Behaviour-changing recall (`on`) is opt-in and NOT enabled in prod.

A5 / WP06 — live validation batch: RECALL_EXPANSION_POLICY flipped `off→on` (2026-10-16, GATE5)

Decision. User-approved GATE5: flip the A5 kill-switch default from `off` to `on` in production and run ONE paid live validation batch to measure the real cell-flip impact of the bounded evidence-expansion.

How the flip was shipped (code default, not env). The Railway CLI on this VM is unauthenticated and there is no token available, and env vars cannot be set on the worker service from here, so — exactly as the A3 enforce-on precedent (`15bbcc0`) — the default was flipped **in code** and deployed via push→auto-deploy: - commit **`f15f8b6`** `feat(WP06/A5): flip RECALL_EXPANSION_POLICY default off->on (GATE5).` - 3 files only: `server/lib/evidence-recall.ts` (constant `DEFAULT_RECALL_EXPANSION_POLICY = "on"` + doc comments), `server/lib/analyzer.ts` (gate-region + import comments updated to the new default; the runtime gate `recallPolicy !== "off" && settings.useBm25Retrieval && evidenceText` is unchanged), and `server/lib/evidence-recall.test.ts` (default-assertion test rewritten to assert default `on` and still prove env `off`/`shadow`/explicit-arg override). Framework-level only — zero per-company/topic strings. - `RECALL_EXPANSION_POLICY` env **still overrides** the default (unset ⇒ resolves to `on`; `=off`/`=shadow` restores prior behaviour with no redeploy). Unrecognised values still fail-safe — now to `on`. - Validation before push: unit 8/8; adjacent suites green (evidence-gate 14/14, evidence-gate-a2 17/17, eligibility-decision-a3 34/34, passage-retrieval 35/35, retrieval-stability 5/5); `tsc --noEmit` 0 new diagnostics in the changed files. Pushed `8a8a4a6..f15f8b6 main`. All 4 Railway services redeployed; health `{"status":"ok","version":"3.0.0"}`.

Policy=on confirmed empirically before the full spend (gitSha signal). Prod has no `RECALL_EXPANSION_POLICY` env var, so the resolved policy on the worker == the deployed code default. There is no pre-batch SHA signal (Railway unauth; health has no SHA; expansion telemetry is console-log-only). The earliest empirical confirmation is the per-company snapshot `analysis_results.results_data[].manifest.gitSha` (written by the worker from `RAILWAY_GIT_COMMIT_SHA`). To cap waste under worker concurrency=10, the batch was enqueued **canary-first**: company 9 alone was queued; it completed healthy (ms=36) and its scores differed from the 1287 baseline (1 verdict flip) — then the remaining 9 were released. On completion, the snapshot gitSha = **`f15f8b6`** on **all 10** companies ⇒ the worker ran the new code ⇒ policy resolved to `on`.

Batch method — corpus-replay off 1287 (deviation: 1284 corpus pruned). The canonical GATE plan was a corpus-replay off baseline **1284**'s frozen corpus, but **1284's and 1285's corpora are pruned** (0 of the snapshot `document_ids` still exist in `documents`; cf. commit `525ac7b` "baseline-1284 corpus pruned"), so a replay off 1284 is impossible. **1287** has a fully

intact frozen corpus (457/457 docs present, all 10 companies) and was itself scored pre-A5-on (gitSha `3990bcd`, A3-shadow ⇒ its pack reflects policy-`off` behaviour), making it the correct isolated baseline. The replay re-scores **1287's exact frozen corpus** with the current deployed code: - replay batch **1288**, score-only, `skipFetch:true`, `sourceBatchId:1287`; the worker loads the frozen corpus from `batch_corpus[1287]` (throws if missing — no live-doc fallback) and forces fresh LLM re-scoring (`freshScoring = !! sourceBatchId`). This isolates the pure A5-on expansion effect from discovery/corpus variance — the same isolation 1284-replay would have given. - source corpus fingerprint `7811721a4ba3941af2772b8f44a067fc32c4b44f7b2a8b7a88e93fd9ad364c8b`; run key `run_a180249bb00d51f1c7bbbbb8074b11a21559bb68a05fab7b427a563bfc86ce85`. - enqueued storage-direct (mirrors `/api/analyze/replay`) because the admin/session auth paths are not reachable from this VM — precedent `validate-enqueue.ts`. - **10/10 completed**, all 10 companies **ms=36** (no measure collapse). Cost **$4.23 / 14.74M tokens** (deepseek-chat, 1,579 calls) vs baseline 1287 $3.89 / 13.52M — the ~+9% token delta is the bounded expansion appending decisive evidence to scoring prompts, as designed.

Cell-flip impact — variant 1288 (policy=on) vs baseline 1287 (policy=off), 10 companies × 36 measures = 360 cells:

Company ID	Company	Yes (baseline 1287)	Yes (variant 1288)	Net	Y→N	N→Y	Total Flips
9	CSX Corporation	5	4	−1	1	0	1
10	MAGNA INTERNATIONAL INC.	12	15	+3	0	3	3
11	THE SOUTHERN COMPANY	3	5	+2	0	2	2
12	S&P GLOBAL INC.	13	14	+1	2	3	5

Company ID	Company	Yes (baseline 1287)	Yes (variant 1288)	Net	Y→N	N→Y	Total Flips
13	MARTIN MARIETTA MATERIALS, INC.	0	1	+1	0	1	1
14	DARDEN RESTAURANTS, INC.	5	4	−1	1	0	1
15	BCE INC.	16	15	−1	1	0	1
16	STRYKER CORPORATION	7	8	+1	0	1	1
17	AMPHENOL CORPORATION	0	0	+0	0	0	0
18	AFLAC INCORPORATED	3	4	+1	0	1	1
ALL	**Aggregate (10 companies)**	**64**	**70**	**+6**	**5**	**11**	**16**

Consistency check (Net == N→Y − Y→N): **PASS** (6 == 11 − 5). Compare artifacts: `/home/ubuntu/a5_gate5_replay.{json,md}` (built by `server/scripts/compare_run_flips.ts 1287 1288`).

Reading. Policy=on produces **16 changed cells (11 No→Yes, 5 Yes→No), net +6 Yes (64→70)**. The bias toward No→Yes (11 vs 5) is the designed

signature of the bounded, append-only expansion: it adds previously-missed decisive passages to the pack (GATE2 measured 537 recoverable misses), lifting more measures over the Yes threshold than it displaces. No measure collapse (all ms=36); no company zeroed that was non-zero in a way inconsistent with the baseline (Amphenol stays 0/0 as in every prior run). Because fresh re-scoring carries run-to-run LLM variance, some of the 5 Y→N and a share of the N→Y are variance rather than pure expansion effect; the net direction and No→Yes skew are the expansion signal.

Pass bar / outcome. ✅ Keep policy **on**. 10/10 clean, ms=36 throughout, expansion effect is coherent and net-positive for recall with no collapse or mass over-flip.

Rollback. `RECALL_EXPANSION_POLICY=off` (or `=shadow`) in the prod env restores prior behaviour with **no redeploy**; or revert `f15f8b6` (3-file default+comments+test change, no schema change), and `ca0939c` to remove the whole A5 feature.

B5 / WP13-C — live validation batch: CONTEXT_COST_POLICY flipped `off→on` (2026-10-17, GATE5)

Decision. User-approved GATE5: flip the B5 kill-switch default from `off` to `on` in production and run ONE paid live corpus-replay validation batch to confirm the context-trim is verdict-invariant (near-zero cell flips) WHILE delivering the designed token/cost reduction.

STEP 1 — offline held-out proof (free, no paid LLM), authoritative for the HARD INVARIANT. Over the frozen corpus of baseline batch **1293** (A5-on, B1-on, B5-off), all **360 cells** (10 companies × 36 measures) were reconstructed as production produces them — REAL `summarizeDocuments` (retrievalV2=false, the workspace-3 setting) + REAL `buildEvidencePacksForCategory` per category + A5 `planBoundedExpansion` then B5 `planContextTrim` at the SAME seam/order as `analyzer.ts` — and `verifyTrimPreservesRecall` asserted on every cell. Result: **invariant holds 360/360, 0 violations** (decisive pack passages ⊆ kept set, recall preserved, identical misses) ⇒ relevant/decisive text is NEVER dropped. Trim fired on **326/360** cells; aggregate pack token reduction **25.39%** (3,920,993 → 2,925,442 pack tokens; 995,551 saved). Per-company pack reduction ranged 6.30% (Aflac) to 32.02% (Magna), each 36/36 invariant-OK. Harness + artifacts committed as `4156742` (`server/scripts/gate_b5_offline_proof.ts`, `gate_b5_out/gate_b5_offline_proof.{json,md}`). This 25.39% is the pack-only first-pass projection; the live batch below measures the true end-to-end prompt delta.

How the flip was shipped (code default, not env). The Railway CLI on this VM is unauthenticated and env vars cannot be set on the services from here, so — exactly as the A3 enforce-on (`15bbcc0`) and A5 GATE5 (`f15f8b6`) precedents — the default was flipped **in code** and deployed via push→auto-

deploy: - commit **8725521** `feat(WP13/B5): flip CONTEXT_COST_POLICY default off->on (GATE5).` - 3 files only: `server/lib/model-cost.ts` (constant `DEFAULT_CONTEXT_COST_POLICY = "on"` + module/kill-switch/ resolver doc comments), `server/lib/analyzer.ts` (gate-region + import comments updated to the new default; the runtime gate `contextCostPolicy !== "off" && settings.useBm25Retrieval && finalEvidence` is unchanged), and `server/lib/model-cost.test.ts` (CC03 rewritten to assert default `on` + prove env `off`/`shadow`/ explicit-arg override; CC07 bad-env now fail-safe to `on`; CC-G3 sets env `off` explicitly for the byte-identical rollback proof). Framework-level only — zero per-company/topic strings. - `CONTEXT_COST_POLICY` env **still overrides** the default (unset ⇒ resolves to `on`; `=off`/`=shadow` restores prior behaviour with no redeploy). Unrecognised values still fail-safe — now to `on`. - Validation before push: model-cost 11/11; adjacent suites green (evidence-recall 8/8, evidence-gate 14/14, evidence-gate-a2 17/17, passage-retrieval 35/35, passage-retrieval-allowlist 4/4, passage-retrieval-empty-corpus 7/7); `tsc --noEmit` 62 before → 62 after, **0 new diagnostics in the changed files**. Pushed `d491c69..8725521 main`. All 4 Railway services (production + sprint-10-preview, app + worker) redeployed with status SUCCESS on **8725521**; health `{"status":"ok","version":"3.0.0"}`.

Policy=on confirmed empirically before the full spend (gitSha signal). Prod has no `CONTEXT_COST_POLICY` env var, so the resolved policy on the worker == the deployed code default. To cap waste under worker concurrency, the batch was enqueued **canary-first**: company 9 alone was queued; it completed healthy (ms=36) and its `pipeline_version` / snapshot `manifest.gitSha` = **8725521** (= the new SHA) ⇒ the worker ran the new code ⇒ policy resolved to `on`. The remaining 9 were then released; the final snapshot gitSha = `8725521` on **all 360 measure_scores rows (all 10 companies)**.

Batch method — corpus-replay off 1293. The replay re-scores **1293's exact frozen corpus** with the current deployed code: - replay batch **1294**, score-only, `skipFetch:true`, `sourceBatchId:1293`; the worker loads the frozen corpus from `batch_corpus[1293]` (throws if missing — no live-doc fallback) and forces fresh LLM re-scoring (`freshScoring = !!sourceBatchId`). Baseline 1293 was scored under B5-off (gitSha `57716af`, pre-flip), so this isolates the pure B5 context-trim effect with A5+B1 on in BOTH runs. - source corpus fingerprint `7811721a4ba3941af2772b8f44a067fc32c4b44f7b2a8b7a88e93fd9ad364c8b`; run key `run_f92811fbf758796070792a3505a07a49e4042dd9ab08cbedc33282c14d8ed999`. - enqueued storage-direct (mirrors `/api/analyze/replay`) because the admin/session auth paths are not reachable from this VM — precedent `a5_replay_enqueue.ts` (reused unchanged; labels via env only). - **10/10 completed**, all 10 companies **ms=36** (no measure collapse).

Token/cost delta — variant 1294 (policy=on) vs baseline 1293 (policy=off), same 1,580 scoring calls (deepseek-chat):

Metric	Baseline 1293 (off)	Variant 1294 (on)	Delta
Prompt tokens	13,937,501	12,221,478	**−1,716,023 (−12.31%)**
Total tokens	14,570,574	12,847,942	**−1,722,632 (−11.82%)**
Cost (USD)	$4.1684	$3.6851	**−$0.4833 (−11.59%)**

Per-company prompt-token reduction was consistent across all 10 companies (10.4% Martin Marietta → 14.4% Stryker) — the trim fired everywhere, not on a subset. The call count is identical (1,580 → 1,580), confirming no structural pipeline change: the saving is purely smaller prompts. This is a **REDUCTION**, the designed outcome of B5 (opposite of A5, which added tokens). The live −12.31% prompt delta is smaller than the offline −25.39% pack projection because the offline figure measured only the evidence-pack portion, whereas the live prompt also carries fixed framework rubric / instruction boilerplate the trim does not touch; both are directionally consistent.

Cell-flip impact — variant 1294 (policy=on) vs baseline 1293 (policy=off), 10 companies × 36 measures = 360 cells:

Company ID	Company	Yes (baseline 1293)	Yes (variant 1294)	Net	Y→N	N→Y	Total Flips
9	CSX Corporation	5	5	+0	0	0	0
10	MAGNA INTERNATIONAL INC.	11	12	+1	2	3	5
11	THE SOUTHERN COMPANY	3	4	+1	0	1	1
12	S&P	14	14	+0	1	1	2

Company ID	Company	Yes (baseline 1293)	Yes (variant 1294)	Net	Y→N	N→Y	Total Flips
	GLOBAL INC.						
13	MARTIN MARIETTA MATERIALS, INC.	0	0	+0	0	0	0
14	DARDEN RESTAURANTS, INC.	4	4	+0	0	0	0
15	BCE INC.	16	18	+2	0	2	2
16	STRYKER CORPORATION	10	9	−1	1	0	1
17	AMPHENOL CORPORATION	0	0	+0	0	0	0
18	AFLAC INCORPORATED	4	4	+0	0	0	0
ALL	**Aggregate (10 companies)**	**67**	**70**	**+3**	**4**	**7**	**11**

Total cells changing Y→N = 4, N→Y = 7. Consistency check (Net == N→Y − Y→N): **PASS** (+3 == 7 − 4). Compare artifacts: `gate_b5_out/b5_flips_1293_vs_1294.{json,md}` (built by `server/scripts/compare_run_flips.ts 1293 1294`).

Reading — the 11 flips are LLM run-to-run noise, not a B5 regression. Three independent lines of evidence: (1) **Below the established noise floor.** Prior paired comparisons on this same 10-company sample — with changes that do NOT touch the scoring surface or (A5) add evidence — recorded Total Flips of A1=21, A2=20, A3=28, B1=19, A5-GATE5=16. B5's **11** is below all of them. (2) **Unstructured.** The 11 flips are scattered across 9 distinct measures and 5 companies (companies 9, 13, 14, 17, 18 had **zero**), max 2 flips on any single measure, and bidirectional (4 Y→N, 7 N→Y) — not the concentrated, one-directional pattern a systematic regression would produce. (3) **Provably no dropped evidence.** STEP 1's invariant proof showed 0 decisive passages trimmed across all 360 cells, so B5 cannot have removed decisive evidence; the residual flips are near-threshold scoring variance from the (recall-preserving) smaller prompt. No measure collapse (all ms=36); the three all-No companies (Martin Marietta / Amphenol, and here Darden's 4) stay stable.

Pass bar / outcome. ✅ Keep policy **on**. Near-zero, sub-noise-floor, unstructured flips WITH a measurable −12.31% prompt-token / −11.59% cost reduction — exactly the B5 success criterion. 10/10 clean, ms=36 throughout, invariant proven 360/360 offline.

Rollback. `CONTEXT_COST_POLICY=off` (or `=shadow`) in the prod env restores the byte-identical pre-B5 pack with **no redeploy**; or revert `8725521` (3-file default+comments+test change, no schema change), and `154ff9b` to remove the whole B5 feature.

B6 / WP14 — capacity experiments: connection-budget-aware worker concurrency (2026-10-18)

Decision. User-approved: "Go ahead with B6 — implement, test and if the test is successful, deploy it to production." B6 delivers WP14's disciplined capacity method — **classify the bottleneck → inventory every DB connection → one-variable capacity matrix → prefer more replicas at lower per-replica concurrency** — plus ONE genuinely dynamic element. It is **NOT** a closed-loop autoscaler and **NOT** a replica-count controller: the replica COUNT stays the user's manual Railway lever; B6 only makes the worker safe/correct at whatever replica count is set, by deriving per-replica job concurrency from the binding DB-connection budget so TOTAL connections stay under the ceiling as replicas scale. Shipped with the kill-switch default `static` (a behavioural no-op), so the deploy carries zero prod risk; enabling `dynamic` remains the user's approval-gated lever.

STEP 1 — bottleneck classification + DB-connection inventory (`gate_b6_out/b6_bottleneck_and_connection_inventory.{md,json}`). - **Binding ceiling, LIVE-confirmed.** The app runs on its OWN Railway-hosted Postgres (project `db04e5b1`), NOT the Abacus platform 25/DB default.

`SHOW max_connections;` = **100**, `SHOW superuser_reserved_connections;` = **3** ⇒ **97 usable** non-superuser connections. The documented “88-vs-97 failure” was a pool bump that pushed the saturated worst-case total toward this 97 ceiling. - **Every production pool inventoried.** `server/db.ts` main drizzle pool `PG_POOL_MAX`=20 (app+worker, eager) + four small lazy gated pools — `llm-usage`/`stage-profile`/`paid-call` = 2 each (app+worker), `run-manifest` = 1 (worker). Per-process worst case: **worker 27, app 26**. App-wide worst case `(appReplicas×26)+(workerReplicas×27)`: 1app+2worker = 80 (OK), **2app+2worker = 106 (BREACH)**, 1app+3worker = 107 (BREACH). Script pools (`server/scripts/*`) do not run inside the live services and are excluded. - **Classification.** DB connections (97 usable) is the **binding** bottleneck; worker concurrency is bounded by pool max (not an independent ceiling); the LLM API is governed separately. **Derived safe budget:** budget 90 (under 97, matches `db.ts` “comfortably under ~90”), reservedOverhead 10, connectionsPerJob 2 (=20/10), HARD_MAX 10 (=today’s static concurrency), `maxSafeReplicas = floor((90−10)/2) = 40`.

STEP 2 — module + wiring + harness. - New pure, DB-free, no-company/topic module **`server/lib/capacity-config.ts`**: `computeDynamicConcurrency = clamp(floor((DB_CONNECTION_BUDGET − reservedOverhead) / replicaCount / connectionsPerJob), 1, HARD_MAX);` the **ceiling invariant** `effectiveConcurrency × replicaCount × connectionsPerJob + reservedOverhead ≤ DB_CONNECTION_BUDGET` is guaranteed by construction for replicaCount ∈ [1, maxSafeReplicas] and exposed via `checkCeilingInvariant()` (beyond maxSafeReplicas it returns false — a documented “cut replicas” signal, never a silent breach — rather than over-claiming “impossible for any N”). - Versioned kill-switch **`CAPACITY_POLICY`** (`static` DEFAULT = current prod/byte-identical / `shadow` = compute + log the budget-derived value but still USE static / `dynamic` = apply the budget-derived value, which can only ever LOWER concurrency as replicas grow, never raise above HARD_MAX). `CAPACITY_POLICY_VERSION=1`, `resolveCapacityPolicy()` fail-safe (explicit > env > default; unrecognized/empty → `static`, total, never throws — mirrors the B5 `resolveContextCostPolicy` pattern). replicaCount reads `REPLICA_COUNT` then `RAILWAY_REPLICA_COUNT` at boot (floor 1). - **Wired ONLY at the worker-concurrency seam** (`server/worker.ts`): `const CAPACITY = resolveEffectiveConcurrency(loadCapacityConfig()); const EFFECTIVE_CONCURRENCY = CAPACITY.effectiveConcurrency;` then `concurrency: EFFECTIVE_CONCURRENCY` on the BullMQ `Worker` (plus the health-check fallback, the adaptive controller’s start value, and the startup log). At default `static`, `EFFECTIVE_CONCURRENCY` resolves to exactly the existing `MAX_CONCURRENT` constant ⇒ byte-identical. Does NOT touch the analyzer/evidence/scoring path and does NOT change any DB pool max size (an approval-gated tuned-matrix decision). - One-variable matrix harness **`server/scripts/gate_b6_capacity_matrix.ts`**: given a budget it prints the computed per-replica concurrency + resulting total DB load + within-

budget flag for a replica-count sweep, so the operator can read the safe operating point for any replica count without touching prod. Sample (default budget 90): r=1→10 (load 30), r=2→10 (50), r=4→10 (90), r=5→8 (90), r=8→5 (90), r=10→4 (90), r=16→2 (74), r=20→2 (90), r=40→1 (90) — every row within budget; r=41 would require cutting replicas.

STEP 3 — tests. `server/lib/capacity-config.test.ts` **13/13 pass**: CAP01 the ceiling invariant holds for replicaCount 1..maxSafeReplicas across a budget sweep (60/75/90/97/120/200); CAP01b beyond maxSafeReplicas the checker flags infeasibility (1×41×2+10 = 92 > 90) rather than hiding it; CAP01c known points (r=1→10,2→10,4→10,8→5,10→4,16→2,20→2,40→1); CAP02 default == `static` and static/shadow USE the static constant (byte-identical) even where dynamic differs; CAP03 dynamic applies the value, only ever lowers, never exceeds HARD_MAX; CAP03b dynamic == static for ≤4 replicas (no change there); CAP04/CAP04b env override (policy + budget knobs + `REPLICA_COUNT`/`RAILWAY_REPLICA_COUNT`); CAP05/CAP05b fail-safe (unrecognized → static; garbage env → defaults); CAP06 determinism; CAP07 purity + shadow-computes-but-does-not-apply. Adjacent **adaptive-concurrency suite 13/13** green (the worker seam shares that controller). `tsc --noEmit`: **62 before → 62 after, 0 new diagnostics in the changed files** (the 62 are pre-existing errors elsewhere in the repo). - **Equivalence (code-path argument).** At the shipped default `static`, the worker's `concurrency` is literally the same `MAX_CONCURRENT` constant it was before, and B6 touches NO analyzer/evidence/scoring file ⇒ the scoring surface is byte-identical ⇒ **Total cells Y→N = 0, N→Y = 0, Total Flips = 0 by construction.** No paid batch is required to prove equivalence.

STEP 4 — deploy. Commit **`deec3a8`** `feat(WP14/B6): connection-budget-aware worker concurrency behind CAPACITY_POLICY (default static=byte-identical)` (4 files: `server/lib/capacity-config.ts`, `server/lib/capacity-config.test.ts`, `server/scripts/gate_b6_capacity_matrix.ts`, `server/worker.ts` seam + `gate_b6_out/` artifacts; no schema/pool/env change). Pushed `30dff06..deec3a8 main`. All **4 Railway services** (production + sprint-10-preview, app + worker) redeployed with status **SUCCESS on `deec3a8`**; health `{"status":"ok","version":"3.0.0"}`. Because the default is `static`, this deploy is a behavioural no-op (safe); enabling `dynamic` in prod remains the user's approval-gated lever.

Cell-flip. Total cells Y→N = 0, N→Y = 0, Total Flips = 0 by construction (no scoring surface mutated at the `static` default).

Rollback. `CAPACITY_POLICY=static` is the default/no-op — nothing to roll back (set `=dynamic` to enable); or revert `deec3a8` (4-file self-contained module + test + harness + worker-seam wiring + gate_b6_out artifacts; no schema/pool/env change).

STEP 5 — production enablement `CAPACITY_POLICY=dynamic` (2026-10-18 ~00:02-00:06 UTC). User-approved: "Switch prod to

CAPACITY_POLICY=dynamic." Set `CAPACITY_POLICY=dynamic` on the **PRODUCTION** environment for BOTH the app and worker services via Railway GraphQL `variableUpsert` (production env only — sprint-10-preview untouched). Each upsert returned `variableUpsert: true`.

- **Env-only change, code unchanged.** This is a variable change, not a code change. Railway auto-redeployed prod app+worker on main HEAD `f41d2af`; `git diff --name-only deec3a8..f41d2af` = ONLY `docs/implementation-run-ledger.{md,docx,pdf}` (the prior docs-only ledger commit) — **no server/code file differs, so the B6 worker/server code running in prod is byte-identical to `deec3a8`.** The tested code == production code invariant holds.
- **Deploy status (observed).** `production worker status=SUCCESS commit=f41d2af branch=main`; `production app status=SUCCESS commit=f41d2af branch=main`.
- **Health (observed).** `{"status":"ok","version":"3.0.0"}`.
- **Variable readback (observed, GraphQL `variables`).** `worker: CAPACITY_POLICY=dynamic`, `app: CAPACITY_POLICY=dynamic`.
- **Runtime policy readback (observed, worker deployment log).** `[Worker] CAPACITY_POLICY=dynamic replicaCount=1 static=4 dynamic=4 effective=4 withinBudget=true maxSafeReplicas=40` — the worker resolved policy=**dynamic** and computed `effective=4` at the current `replicaCount=1`, which is **equal to the `static` value (4)**, so concurrency is unchanged right now. `withinBudget=true` confirms the ceiling invariant holds. Dynamic can only LOWER per-replica concurrency once replicas scale (≥5), never raise it ⇒ ceiling-safe by construction.
- **Cell-flip. Total cells Y→N = 0, N→Y = 0, Total Flips = 0** by construction — a worker-concurrency change cannot alter scoring; analyzer/evidence/scoring surface untouched; no paid batch run (none needed — the runtime readback shows dynamic == static at the current replica count).
- **Rollback.** Set `CAPACITY_POLICY=static` (or unset) on the prod app+worker to restore the byte-identical no-op default — no redeploy of code required.

A9 / WP15 — truthful publication & release-evidence governance (2026-10-18)

Decision. Backlog step 9 of 10 under the standing rule "implement → test → keep if it works / roll back if not → deploy to production on passing tests." A9 delivers WP15: make every exported assessment *honestly* describe what it is (framework version, source scope, coverage/research status, evidence, provenance), make a reviewer able to tell an UNCHANGED score from a quietly-INCOMPLETE one, make published history immutable with explicit revisions, and make release evidence raw and non-deduped so nothing can

hide. It is pure reporting/governance — it touches **no analyzer/evidence/scoring path** — so it ships behind a kill-switch whose default is a non-mutating passthrough, carrying zero prod risk.

STEP 1 — module. New pure, DB-free, no-company/topic module **`server/lib/truthful-publication.ts`** (+ `server/lib/truthful-publication.test.ts`). Reuses existing provenance primitives rather than inventing parallel ones: `shared/verdict.ts` (`normalizeVerdict/isMet` — the one canonical verdict rule), `shared/research-state.ts` (`ResearchStatus/summarizeResearchStatus/researchStatusCsvColumns` — CV01–CV07), `server/lib/run-manifest.ts` (`RunManifest/hashObject/canonicalize`). - **(a) One canonical publication record.** `buildPublicationRecord(input)` is the single accessor every reader surface derives from (API, UI, CSV, Word, PDF), so verdict / evidence / research-status can never diverge between surfaces (PU01, CV07). The record links the assessment to framework version + hash, source scope + as-of (`PublicationScope`), coverage/research status (`ResearchStatus`), ordered evidence IDs, source document + link, and run provenance (`runId/runMode/manifestHash/codeTreeHash/gitSha` + `reproducibilityLimits` carried verbatim from the run manifest). - **(b) Incomplete ≠ comprehensive.** `assertTruthfulCompleteness(record, claimedComprehensive)` returns a violation when an incomplete assessment is claimed comprehensive; `comprehensive` is TRUE only when research is comprehensive with no unresolved essential gaps. A quietly-incomplete cell is surfaced as incomplete, never rendered as comprehensive. - **(c) Immutable snapshots + explicit revisions.** `publishSnapshot()` deep-freezes a content-hash-anchored snapshot (a later mutation of the caller's objects cannot change history). `reviseSnapshot(previous, …, reason)` creates an EXPLICIT new revision (increments `revisionNumber`, links `revisionOf`, requires a reason — throws on empty) and leaves the original untouched; a correction is always a new, dated, linked revision, never a silent recompute. `verifySnapshotIntegrity()` re-hashes and compares. - **(d) Separate release record types.** `candidate-review | approval | merge | deploy | validation | acceptance`. `classifyCommitAsReleaseEvidence()` returns a HISTORICAL `merge` record — a commit message (even one saying "approved/LGTM") is never promoted to approval. `recordApproval()` throws if asked to approve from a commit message or without a named approver + reference. `preserveFeatureBranchText()` keeps embedded "feature-branch-only / do-not-deploy" text as dated history. `currentReleaseStatus()` ignores historical records, so historical text can never be read as current deployment status. - **(e) Release-gate evidence — raw, non-deduped.** `evaluateReleaseGate(input)` preserves the RAW lists (every test occurrence incl. skips, every TS diagnostic with multiplicity, mutation checks, packaging) verbatim and presents a dedup SUMMARY ALONGSIDE — never instead of — them (relevant text never dropped). The gate FAILS (never an exit-0 skip) on: any failed test; a skip with no explicit reason; TS

diagnostics over the accepted baseline (by RAW count); **missing Chromium (PU04)**; **a mutation guard that did not fail as expected (PU05)**; **a missing required subprocess/worker asset (PU06)**. `reconcileRuntimeWithTested({deployedSha, testedSha})` enforces deployed SHA == tested SHA (PU03, short/long-SHA equivalence). - **(f) WP23 measurement baseline.** `MeasurementBaseline` keyed by framework/topic/language/sector, with `incompleteResearchRate` tracked SEPARATELY from positive precision/recall so a global average can't hide a small-issuer / non-English subgroup gap. Serializes/round-trips canonically. - **Kill-switch.** `PUBLICATION_POLICY` — `passthrough` (DEFAULT, non-mutating, byte-identical to today) / `annotate` (APPENDS an additive governance block + research-status CSV columns; existing fields never reordered or changed) / `shadow` (also non-mutating on the reader surface). `PUBLICATION_POLICY_VERSION=1`; `resolvePublicationPolicy()` fail-safe (explicit > env > default; unrecognised/empty → `passthrough`, total, never throws — a typo can never silently change what a reader sees).

STEP 2 — cross-surface routing. The TS accessor is the single canonical source of the published verdict and status. The Python CSV exporters (`server/scripts/export_ai_csv_from_snapshot.py:norm_verdict`) already implement the identical rule as `shared/verdict.ts:normalizeVerdict` (non-string→"No", blank→"No", "partial"→"No", else verbatim); A9 adds a committed **Python/TS parity test** (PU01c) that pins the two cell-for-cell so a CSV/Word/PDF export can never show a different verdict from the API/UI. No invasive exporter rewrite was made — that preserves the byte-identical default and avoids prod risk.

STEP 3 — tests. `server/lib/truthful-publication.test.ts` **18/18 pass**: PU01 cross-surface consistency, PU01b incomplete≠comprehensive, PU01c Python/TS verdict parity; PU02 immutable snapshot + explicit linked revision (empty reason refused); PU03 SHA reconcile; PU04 missing-Chromium fail + PU04b unreasoned-skip fail; PU05 parent-cancellation mutation sensitivity (blind spot → fail); PU06 worker-asset packaging; TS-baseline gate (raw count); RAW-EVIDENCE preserved (duplicate test/diagnostic occurrences kept, dedup summary alongside); release-record types (commit≠approval, feature-branch=history); KILL-SWITCH default passthrough byte-identical (exactly the 5 base CSV columns, no governance block; shadow also non-mutating); NON-MUTATION (input never mutated, evidenceIds copied not aliased, verdict uses only the existing normalizeVerdict — Partial→No is the existing canonical rule, not a new A9 transform); RESOLVER fail-safe; ROUND-TRIP serialization + stable hash; WP23 subgroup baseline. Legacy suites green: capacity-config 13/13, run-manifest 7/7, snapshot-durability 12/12. `tsc --noEmit`: **62 before → 62 after, 0 new diagnostics in the changed files** (the 62 are pre-existing errors elsewhere in `server/scripts`).

Cell-flip (code-path argument). A9 is reporting/governance only. At the shipped default `passthrough`: no published-score surface is mutated — the `verdict` field is produced solely by the pre-existing canonical `normalizeVerdict` every surface already applies, and `passthrough` emits no new or changed reader field (the governance block / research-status columns appear only under the opt-in `annotate`). No analyzer, evidence, or scoring file is touched. Therefore **Total cells Y→N = 0, N→Y = 0, Total Flips = 0 by construction** — no paid batch is required to prove equivalence.

STEP 4 — deploy. Commit **`8aebe6b`** `feat(WP15/A9): truthful publication & release-evidence governance behind PUBLICATION_POLICY (default passthrough=byte-identical)` (2 files: `server/lib/truthful-publication.ts`, `server/lib/truthful-publication.test.ts`; no schema/pool/env/exporter change). Pushed `fd0625f..8aebe6b main`. All **4 Railway services** (production + sprint-10-preview, app + worker) redeployed with status **SUCCESS on `8aebe6b`** (full `8aebe6bb6dfbdbd26e7c563b8ee34af0b55fd13a`); prod health `{"status":"ok","version":"3.0.0"}`. - **Deployment-target verification.** deployed SHA == tested SHA on ALL 4 services (prefix match of the pushed commit `8aebe6b`). Because the default is `passthrough`, this deploy is a behavioural no-op (safe); surfacing governance via `annotate` remains an opt-in lever.

Rollback. `PUBLICATION_POLICY=passthrough` is the default/no-op — nothing to roll back (set `=annotate` to surface governance columns; no code redeploy if the env is settable); or revert `8aebe6b` (2-file self-contained module + test; no schema/pool/env/exporter change).